2026-2027

ENGLISH FOR SCIENCE I

(V-LANG-650)

BAB2 COMPUTER SCIENCE

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UNIVERSITÉ DE MONS

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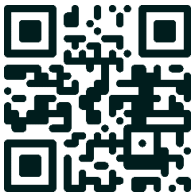
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Essential websites

OXFORD ENGLISH ADVANCED DICTIONARY	LINGUEE	HOWJSAY	SKELL
Dictionary & pronunciation: general)	*Specialised dictionary with real-life examples*	*Pronunciation: scientific & specialised words*	*Collocations*
			

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GUIDELINES ABOUT THE COURSE

I – GENERAL INTRODUCTION

Course Title: ENGLISH FOR SCIENCE

Credits (UE): 5 ECTS

Prerequisite: None

Aim: B1

Teaching methods: 48h in-class activities + Moodle activities

Learning material:

- Student coursebook
- Grammar book
- Moodle activities

Learning outcomes:

At the end of this course, students will be capable to

- use in context basic English grammar notions;
- extract information from science-oriented English texts;
- communicate in English, in writing and orally, on science-related topics;
- work autonomously on audio documents (B1 level);
- build vocabulary linked to science;
- put into practice the basic rules of English pronunciation;
- create science-oriented works linked to their field of study.

English for Science, part 1 aims to teach you basic scientific English concepts and vocabulary and offers a thorough revision of English tenses.

The course consists of a mix of theoretical and practical activities (exercises and project) and Moodle activities.

The course content is divided into five main units: (1) *Defining, Describing and Qualifying*, (2) *Purpose and Process*, (3) *Measuring and Quantifying*, (4) *Frequency* and (5) *Linking Your Ideas*.

At the beginning of the year, you will have to take an eLAO test. If you score at level A1 or A2, we invite you to attend tutoring sessions ("cours inter-facultaires").

II – COURSE GUIDELINES

1. Contacting the teacher

Address your teacher **only** in English, inside and outside class.

Ask all course related questions on the Questions forum in English (Front Page section) on Moodle.

Do not send e-mail unless your question or message contains strictly confidential information. Emails written in French will not be answered.

Asking questions on the forum serves a double purpose. First, you will inform your colleagues of any relevant matters that affect them as well, such as your absences. Second, you help me save time and contribute to the needs of others by asking on the forum, as others are likely to have the same question. Lastly, you will get a faster reply via the forum, as I check my forum messages before my e-mail. This is my method to prioritize students' communication over others in my workflow.

2. Participation

All work, including homework, Moodle activities and active class participation (not just attendance) is **highly encouraged**. Practice makes perfect!

3. Moodle activities

Homework

Each unit contains individual and study circle homework and in-class tasks. The homework should be completed before the next class to ensure logical progression between units (certain activities such as 'Test Yourself!' can be completed later if necessary to revise for the exams).

Self-study

All self-study activities are available from Week 2 and need to be completed by December 13 23:59.

The full completion of Moodle activities (see completion progress on Moodle) is required to obtain the bonus.

4. Plagiarism/AI awareness

The University of Mons has zero tolerance for **plagiarism**. The University of Mons considers the use of **AI** (ChatGPT, ClaudeAI, Perpexity, etc.) as academic dishonesty or fraud if not cited accordingly (for short excerpts, in the same way as you would do it for any other work). Make sure to properly cite AND paraphrase any work you submit. Please read the documentation on the use of AI at the University of Mons.

The citation standard in this course is APA7: https://apastyle.apa.org/

III – COURSE ASSESSMENT MODALITIES

TERM 1: 30 points out of 100 (English for Science final grade)

- Vocabulary test: 50/100
- Grammar test: 50/100
- Term 1 Moodle activities: 5/100 bonus if 100% completed

One grammar and vocabulary exemption test will take place during term 1. If you get 12/20 or above in grammar or vocabulary, you will get an exemption for that part. If you get less than 12/20 for either of these parts, you must take the corresponding exam during the session.

You are required to attend **at least** 60% of the lessons (= 7/12 sessions) in order to be allowed to take the exemption test, regardless of the reason for the absence.

TERM 2: 70 points out of 100 (English for Science final grade)

- Vocabulary test: 15/100
- Grammar test: 25/100
- Final oral presentation about the project: 60/100
- Term 2 Moodle activities: 5/100 bonus if 100% completed

One grammar and vocabulary exemption test will take place during term 2. If you get 12/20 or above in grammar or vocabulary, you will get an exemption for that part. If you get less than 12/20 for either of these parts, you must take the corresponding exam during the session.

You are required to attend **at least** 60% of the lessons (=6/10 sessions) in order to be allowed to take the exemption test, regardless of the reason for the absence.

TERM 3: 100 points out of 100 (English for Science final grade)

Term 1 retakes

- Vocabulary: 50 points
- Grammar: 50 points

Term 2 retakes

- Vocabulary test: 15 points
- Grammar test: 25 points
- Final oral presentation about the project: 60 points

IV – PROJECT

1. Project overview

What?

You will work on a project throughout the year and present it during the oral exam in June.

The topic must be specific enough: an invention, a technique, a health programme, a pathology, a machine, a research project, …

Your project must:

Part 1 (Term 1)

1. be identifiable, with nuances
2. have a specific purpose and involve one or several processes, i.e., why was it created and how does it work?
3. be sizeable, measurable (basic data)

4. involve recurring patterns and/or phenomena, be statistically verifiable (e.g., disease recurrence, project evolution, etc.) (data, tables, graphs)
5. be presented with a clear structure, stating its origin and effect, among other elements.

Part 2 (Term 1 + Term 2)

6. include all the elements listed for Term 1 (recap of Term 1 at the beg. of Term 2)
7. include several hypotheses and interpretations
8. contain specific concepts that can be thoroughly defined
9. be drafted in an impersonal, formal style in writing and be presented a less formal/more personal style in your speeches
10. include comparisons of concepts (show how similar/different your project is from other similar research topics) (data, tables, graphs)

How to start?

State of the art: look up popular science articles and see how trendy your topic is (which means you will find plenty of articles and you will not thus struggle to complete your project!). **Do not come up with an idea out of the blue without doing actual research**. Your project must be based on new, interesting research and be widely developed in the scientific press (popularised press mainly, and academic).

The project topic choice is on a first-come, first-served basis on Moodle. One project per person. Please ask if you are not sure about your project proposal.

You must finalise your choice of topic and share the hyperlink to your shared PowerPoint by 24th September 2026 -11.59 PM.

When?

- 15th Nov. 2026 – 11.59 PM:
  - Current research : selection of articles 1-5 (Shared Excel Sheet)
  - Assessment in class by your peers of the suitability of your articles: bring a paper version of your articles in class (2 pages per sheet, double-sided = 4 pages on 1 sheet of paper).
- 29th Nov. 2026 - 11.59 PM: First PPT Project checkpoint
  - Current research: selection of articles 1-5 (Shared Excel Sheet)
  - PPT: articles 1-2
- 21st March 2027 - 11.59 PM: Second Project checkpoint
  - Current research : submission of your final list of articles 1-5 (Shared Excel Sheet). *No late submission will be considered. 3/60 will be deducted per current research article that does not meet the criteria.*
- 11th April 2027- 11.59 PM:
  - PPT: articles 1-4

- Two days before the exam: submit your final PowerPoint presentation on Moodle (use PowerPoint template + submit on final assignment)

2. Analysis table document

Before inserting current research information (5 articles – see below) into the main body of your Powerpoint, analyse each article with the **analysis table document**. This analysis table is based on the IMRad format. It will help you synthesize the information efficiently and therefore extract valuable information to create synthetic slides.

3. Project presentation and Powerpoint instructions

In class:

- Short presentations and group discussions throughout the year to prepare you for the final presentation
    - use the template provided in the Project Exam: info & deadlines section
    - stick to the same Powerpoint for the whole year, adding information as you proceed
    - share the file with me before the first presentation in September

Oral presentation (exam):

You need to prepare a PowerPoint presentation.

Duration: actual presentation: 8' per person + 2' Q&A

Content:

There will be five parts to your presentation (the information inside each section can be laid out in a personalised manner for your final presentation, as long as each information item is included) + Q/A session + feedback.

a. **Overview/ Rationale** (one minute max): What is your topic? Why did you choose this subject? Why is it interesting?

b. **General information** (min. 3-4 sources + min. one audio/video): includes all the information gathered through the year via the in-class activities:

- *Defining, describing and qualifying* – general object description (with qualifiers)
- *Purpose & process* – origin and use of objects (can be same objects as above or new ones)
- *Measuring & quantifying* – general project data
- *Frequency* (+ *qualifiying*) – graph description: thorough project data
- *Terminology: compound forms* – explain at least two key (complex) concepts found in your documents.
- *Comparing*: compare two concepts (one from your project and another found in literature, on the web, etc.).

More information can be added to understand the general context of the project.

c. **Current research** about the topic (5 articles: one research article and 4 magazine/news articles of minimum 500 words per article and from at least 2 different sources)

- Give an overview of the link between articles.
- Be prepared to present all articles but note that you will not have time to cover all of them. Your teacher chooses which article(s) you will talk about just before the presentation.
- *Hypothesizing* (conditionals and modals): explain at least two hypotheses found in your studies.

All the sources must be about recent findings. Please do not select encyclopedic articles for this section (= general information)

d. **Take home message/conclusion**: what should we remember from your presentation (one recap slide - leave it on during Q/A)

e. **References**

- Cite all your sources in APA7 (reference section in the template) on each slide (with the use of numbers in the bottom banner) and at the end of the Powerpoint. This is the most reliable way of citing sources on a Powerpoint.
- The relevant use of each source in your project is left for you to decide depending on your needs (for example: to explain a complex concept or term, analyse the origin of the project, understand the implications of the project, highlight the recent evolution of the project, etc.). You must be capable of explaining why you chose a specific reference to support your claims and why it is a reliable source, according to academic standards.
- If you hesitate about the reliability of a source, please ask.

f. **Question/Answer session:**

(non-exhaustive list)

- Which article was the most difficult? Why?
- What is your favourite article and why? How is it different from the others? What does it talk about?
- What was the most difficult task while doing your research and why?
- How can you apply this knowledge in daily life?
- What did you want to transmit to other people talking about this subject?
- Did you learn more about it? If yes, what did you learn?
- Would you like to continue to work on that subject later? If yes, which Master's specialization course(s) are you looking forward to taking?
- If you had to choose another subject, what would you choose?
- What do you think about the future of your topic?

g. - feedback about the presentation (by the audience - other students)
- language feedback (by the teacher)

Visual aid:

- Make sure to include many figures (graphs, diagrams, etc.) to illustrate general information and current research. Otherwise, stick to KEY WORDS.
- NO reading. You may have cards with keywords to help but strictly NO text reading.
- Slides: PowerPoint, LaTeX...
- Handout: IF applicable, give a handout (with tables/graphs that are hard to read, ...)

V – COURSE MATERIALS & RESOURCES

Compulsory course materials

- Coursebooks (1 per term)
- Oxford English Grammar Course (intermediate - with answers). 2 versions exist:

  . the older version: ISBN: 978-0-19-442082-2

  . the new version: ISBN: 978-0-19-441487-6 (5% discount at Scientia for UMONS students)

They are exactly the same content wise, same page numbers, ...

All students must have access to both these texts during class, either in the form of textbook, printed copy, or electronic copy. Either the current edition or the previous edition may be used. Classes will usually consist of some review of these materials, as well as listening practice, writing exercises, quizzes, and discussion.

Additional course materials

- **Quizlet**

  Procedure to join Quizlet

  1. Go to the Quizlet website

  2. Join Quizlet
     Sign up
     Fill in
     - username: one word only, no spaces
     - email address
     - password
     - tick 'I agree'

  3. Go to your email inbox and confirm

  4. Come back to this page and click on

     a. Quizlet FS class term 1

     b. Quizlet FS class term 2

  5. Settings

     - Choose profile picture

- Language : choose English UK
- Email alerts: untick all

You can now practise most of the basic compulsory vocabulary required in this course.

- English resource Bank : Learn online; study and practise speaking English in Mons or abroad; watch English programmes; use academic and business glossaries…
- Grammar Bank: Learn and practise grammar online (intranet)
- Blattes, S., Jans, V., & Upjohn, J. (2013). Minimum competence in Scientific English. EDP Sciences. https://doi.org/10.1051/978-2-7598-0989-9
- Upjohn, J., Hay, J., & Colle, P. (2006). Minimum competence in Medical English. EDP Sciences.

Other useful resources

1. Pronunciation tools

- **Howjsay**: Use Howjsay to listen to the pronunciation of any word, including specific, scientific, and technical terms. This tool is particularly useful for mastering difficult pronunciations.

- **Contextual Listening (YouGlish/Youtube)**: Utilize YouGlish to hear how specific words are used in different contexts and accents on YouTube. This can help you understand the nuances of pronunciation and usage. If you cannot find the word with this tool, simply type it on Youtube and check reliable videos from conferences, etc. (note: **avoid** at all costs AI pronunciation videos such as this one)

- **Dictionaries**: Refer to dictionaries such as Merriam-Webster and Cambridge to check the phonetic transcription of words. This will help you understand the correct pronunciation and improve your phonetic skills.

2. Dictionaries

- Wordlink. This fantastic tool can link most webpages word-by-word to online dictionaries. Just copy and paste the URL of your text, specify the webpage language and target language. Then, select the dictionary you want to use :

  . ***monolingual*** (target language: to English): Oxford Learner's Dictionary or WordReference

  . ***bilingual*** (target language: to French): WordReference, Linguee, Larousse

  Copy and paste a URL into the box above, specify the language, and click 'Go' to a display copy of the page with all words linked to dictionary lookup.

- Merriam Webster (monolingual with pronunciation)
- Macmillandictionary.com
- Wordreference (bilingual)
- The Free Dictionary (bilingual)
- Oxford Online Dictionary (monolingual with pronunciation)
- Linguee (translation engine)
- Tradooit (bilingual concordancer)
- Cambridge Online Dictionary (monolingual, bilingual, speciality areas, …)
- Collins Online Dictionary (monolingual)

- Just The Word (monolingual concordancer)

3. Magazines and Press (popular science articles) [SOURCES FOR YOUR PROJECT]

Before looking up information on any website, check if the University of Mons has paid access through the UMONS resources (especially Press Reader - look under 'Bases de données'). The University of Mons has institutional access to all these magazines:

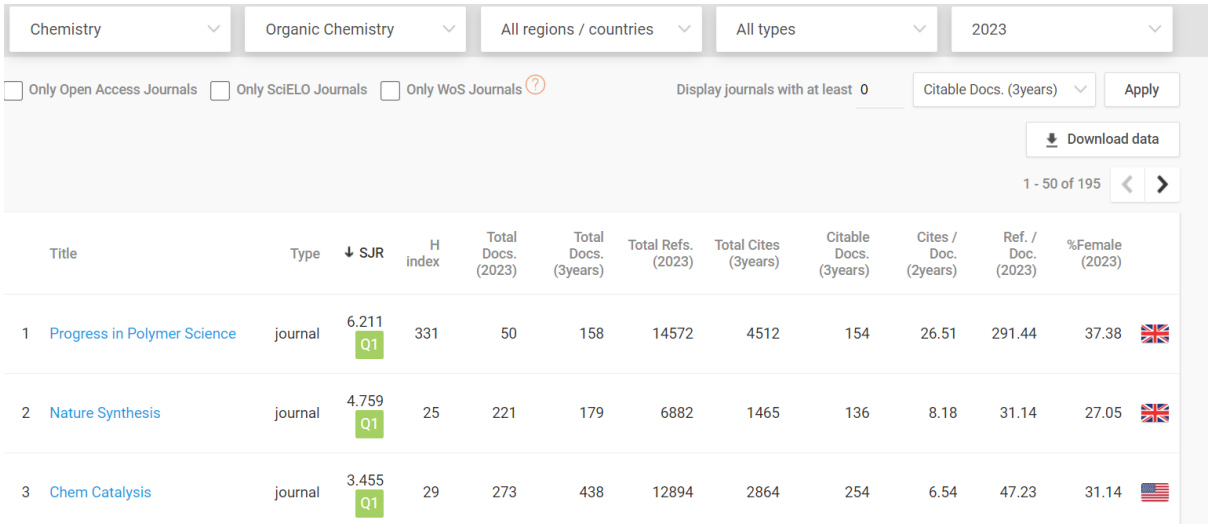
- New Scientist
- Scientific American
- Science News
- BBC Science Focus
- Cosmos
- Smithsonian

Free resources online (non-exhaustive list)

- ScienceDaily
- Phys.org
- Science Mag
- Live Science
- EurekAlert
- E-science
- BBC Science and Environment
- ABC Science
- NASA Science
- Science News for Students
- Quantum Magazine

4. Scientific resources (journals) [SOURCES FOR YOUR PROJECT]

The University of Mons has an institutional access to many journals. Identify a journal from your field of study (you can use this website – start by selecting the subject area (e.g. Chemistry) and subject category if you wish (e.g., Organic chemistry), select a journal and find a scientific paper that corresponds to your project topic.



	Title	Type	SJR	H index	Total Docs. (2023)	Total Docs. (3years)	Total Refs. (2023)	Total Cites (3years)	Citable Docs. (3years)	Cites / Doc. (2years)	Ref. / Doc. (2023)	%Female (2023)
1	Progress in Polymer Science	journal	6.211 Q1	331	50	158	14572	4512	154	26.51	291.44	37.38
2	Nature Synthesis	journal	4.759 Q1	25	221	179	6882	1465	136	8.18	31.14	27.05
3	Chem Catalysis	journal	3.455 Q1	29	273	438	12894	2864	254	6.54	47.23	31.14

Note that you cannot choose a review paper, which is a different type of scientific publication (slightly more complex and especially longer than a conventional research paper). If you do not know the difference between a research paper and a review paper, go to this page.

5. General and scientific audiovisual resources

General

- Breaking News English
- BBC Learning English
- *Voice of America - Learn American English with VOA Learning English*
- English Central

Scientific

- TED-Ed (short videos on various subjects including science, 2 to 7 mins with or without subtitles)
- Domain of Science
- Kurzgesagt (short videos on various subjects including science, 2 to 7 mins with or without subtitles)
- Ted Talks
- Inside Science
- Science in Action
- Science Friday
- The Open University
- Science
- The Life Scientific
- PBS Space time (for physicists, advanced level)

6. APA style (referencing guidelines)

- Reference list: https://apastyle.apa.org/instructional-aids/creating-reference-list.pdf
- In-text citations: *In-text citations*. (s. d.). https://apastyle.apa.org. https://apastyle.apa.org/style-grammar-guidelines/citations
- Free APA7 citation generator: *APA7 Citation Generator : CiteFast automatically formats citations in APA 7th Edition*. (s. d.). Cite Fast. https://www.citefast.com/?s=APA7#_Webpage

7. Proofreading, paraphrasing tools & improving your English with AI

- DeepL write: AI-powered writing companion
- Writefull
- Quillbot
- LanguageTool
- Online AI Editor & Grammar Checker - Wordvice AI
- Lengusa | Write better English!
- Ludwig • Find your English sentence
- Sentence Stack
- Grammarly

- "SpellCheckPlus" Online Spelling and Grammar Checker for English as a Second Language.

VI – SAVE THE DATES!

- 16th September (23:59): complete the eLAO test

- 24th September (23:59): your project topic needs to be submitted

- 15th November (23:59): submit your selection of articles 1-5 (shared Excel sheet) + bring a paper version of your articles in class to assess their suitability.

- 29th November (23:59): 1st PPT project checkpoint: presentation of your articles 1 and 2 to other students using your PPT.

- 6th December (23:59): all Moodle activities should be completed to get 1 bonus point (1/20 of the term 1 final grade)

- 14th December: exemption tests in grammar and vocabulary

- 21st March (23:59): submission of your final list of articles 1-5 (shared Excel sheet)

- 11th April (23:59): second PPT project checkpoint: presentation of your articles 1-4 to other students using your PPT.

- 2nd May (23:59): all Moodle activities should be completed to get 1 bonus point (1/20 of the term 2 final grade)

- 10th May: exemption tests in grammar and vocabulary

STARTER

I – WAYS TO IMPROVE YOUR ENGLISH

In groups of 3-4:

1. Discuss ways to improve your language skills (Speaking, Writing, Listening, Reading and Vocabulary), and write down your ideas below:

2. Strike a contract with yourself: decide on 3 changes in your life to learn English more actively and write them down below:

   -
   -
   -

1. 10 RESOURCES TO IMPROVE YOUR LISTENING SKILLS & SPOKEN ENGLISH IMMERSIVELY

1) Travelling/living abroad[i]: this is based on the idea that the active use of a foreign language in a variety of real-life circumstances and contexts is the best way to learn it. This method has several advantages. It accelerates learning and helps improve oral expression, promotes cognitive growth, improves memory and problem-solving ability. It can also help increase cultural understanding and empathy, helping students have a greater awareness and respect for other cultures. Read this experience to learn further on the topic.

2) AI Applications: To enhance your English language skills independently, consider utilizing various AI applications. Some of these apps, such as Pronounce and ELSA Speak, offer limited free plans. For a comprehensive list of AI pronunciation tools, you can visit this website.

3) Original Version with English Subtitles: Make a conscious effort to watch your favourite series and movies in their original version with English subtitles. This method is effective because it allows you to hear the natural flow of the language while simultaneously reading the text. Link to read further on the topic.

4) English-Speaking TV Channels: Watching English-speaking TV channels can also be beneficial. Many of these channels have excellent apps and YouTube channels, such as the BBC News App available on Google Play and the Apple Store, and the BBC News YouTube channel.

5) YouTube Videos: You can also watch virtually any video in English on YouTube, including entertaining ones that make learning English enjoyable.

6) **Music in English**: Listening to music in English is another great way to improve your skills. For active listening practice, you can use platforms such as LyricsTraining.

7) **Video Games**: Playing video games in English, especially massively multiplayer online games (MMOs) like Fortnite, World of Warcraft (WoW), and League of Legends (LoL), can be particularly helpful. Games with rich storylines, such as Zelda, Horizon, Grand Theft Auto (GTA), and The Last of Us, also provide immersive language experiences. For additional resources, you can explore the website Real English for Gamers.

8) **Podcasts**: Listen to podcasts on Spotify, Deezer, and Apple Music. Many podcasts come with transcripts, which can aid in understanding and learning new vocabulary.

9) **Language Exchange/Events**: Attend English events or find language exchange opportunities in Mons through platforms like My Language Exchange. These events provide a great way to practice speaking with native speakers and other learners. *The Bookshop* in Mons offer such events free of charge. My dear colleague Amy also organises events through her Facebook group English Evenings.

10) **Museums**: Visit museums that offer audio guides in English. This can help you improve your listening skills while learning about various subjects.

2. 17 TIPS TO IMPROVE YOUR PRONUNCIATION

1) You should be aware of the **movement of your lips, tongue, and throat**, and it's helpful to check your progress in the mirror. You might begin with this video.

2) You can **record yourself using your phone's voice texting feature**, such as WhatsApp's own conversation.

3) **Practicing shadowing with transcripts** from TED Talks, audiobooks, podcasts, videos, songs, and other sources can be beneficial. Shadowing is a technique where you listen to a native speaker and try to mimic their speech as closely as possible, including their intonation, rhythm, and pronunciation. You can also create your own transcripts with the free online text-to-speech AI tool Luvvoice in order to practise your pronunciation (which is very useful to prepare for a presentation). Combine it with Reading Coach (see 5.).

4) **Imitating common phrases and words** used in casual conversations, like "oh I donno" or "hi whatcha doing?" can be a good start. You might begin with this video.

5) **Reading aloud** regularly helps you strengthen your mouth muscles. You might begin with Reading Coach.

6) How to use it**:** select 'add your own passage' and paste the text you want to practise reading. Then, allow access to your microphone for recording. Read out loud, making sure to articulate and speak loudly enough and slowly. Practice individual words to improve your pronunciation.

7) **Emphasizing key phrases** can improve your clarity and expression.

8) Focusing on Erreur ! Source du renvoi introuvable. enhances your pronunciation.

9) **Comparing different sounds and words**, especially minimal pairs, can refine your listening and speaking skills.

10) **Practicing the seven intonation patterns** in your speech is extremely important. One example with this Friends video.

11) **Working on contractions** can help you sound more natural in conversations. You might begin with these two videos: affirmative contractions and negative contractions.

12) **Singing along to slow songs** by artists such as The Beatles or Frank Sinatra can be enjoyable, as new songs tend to be fast.

13) **Trying tongue twisters** can be a fun way to improve your pronunciation and articulation.

14) **Comparing English sounds with French** can help you understand the differences and improve your pronunciation. Have a look at these eight points of comparison.

15) **Remaining consistent with British English** in terms of pronunciation, vocabulary, and grammar is important (see this video). If you're not confident with another form of English, it's best not to improvise. There are exceptions for students who have studied in SHAPE or lived in the US, Australia, etc.

16) **Accepting feedback** from friends or teachers is crucial for improvement. Remember, it's nothing personal!

17) **Practising regularly** is key because practice makes perfect.

18) **EngVid pronunciation exercises** can be a helpful resource for improving your pronunciation.

II – STUDENTS' INTRODUCTIONS

1. Whole class: Get up, small talk to min. 4-5 classmates, and find the most successful strategies. Introduce yourselves:

- What is your name, how old are you & where are you from?
- Tell me something about your hobbies and interests
- What do you study and why?
- What is your favourite scientific topic/course?
- What are your career plans?
- Experience of learning English: how many years? How do you immerse yourself?
- When did you last use English? With whom? In which situation? How did you feel before, during and after communicating in English?
- Have you ever been to an English-speaking country? Which one?
- What do you expect from this course?

2. Plenary

3. Judgmental Language: give examples from your personal experience.

What is "good/bad" or "(in)correct" English like? Is that question even acceptable?

Space

Aliens on low-oxygen worlds may never discover fire

Alex Wilkins



YURIY MAZUR/SHUTTERSTOCK

Some planets could host complex life, but nothing will burn there

LIFE on a planet with low levels of oxygen might never be able to develop technology because combustion would be impossible. This bottleneck for creating advanced civilisations may also help explain why we have yet to observe life elsewhere in the universe.

The development of more sophisticated technology on Earth hinged on fire, also known as combustion, a chemical reaction that uses oxygen to generate large amounts of heat.

This is crucial for extracting metals from ore, say, which almost all advanced human technology relies on. Lab experiments have found that combustion can't fully occur in atmospheres with oxygen levels below about 18 per cent.

Now, Adam Frank at the University of Rochester in New York and Amedeo Balbi at the University of Rome in Italy suggest that this might be a bottleneck for whether alien species can develop technology.

"You may have enough oxygen in the atmosphere of an exoplanet to have complex multicellular life, but you may not have enough oxygen to start combustion," says Balbi.

We don't know the exact balance of geological and biological processes that lead to an atmosphere with this level of oxygen because we only have Earth as an example. "Oxygen in the atmosphere is also produced by photosynthetic activity, so that's why it gets complicated, because it's not just geology, it's also biology," says Balbi.

18%

Amount of oxygen needed in atmosphere for full combustion

There is also an upper limit to a useful amount of oxygen. At about 30 per cent and above, depending on the air's moisture content, the chance of widespread fire killing life forms would become high as combustion becomes so easy, says Frank. Astronomers are looking for any type of life in the universe, but the search for advanced aliens might be guided by seeking planets with atmospheres that are between 18.5 and 21 per cent oxygen, the optimum for useful combustion (arXiv, doi.org/kptb).

For now, though, detecting atmospheric oxygen is beyond the abilities of our telescopes for almost all known exoplanets. "We're just getting started looking for the atmospheres of terrestrial planets," says Frank. "This is the exciting thing – we're at the frontier now."

Combustion as a bottleneck for technology makes sense if life emerges how it did on Earth, but might not apply if life elsewhere evolved differently, says Ingo Waldmann at Imperial College London. "That was the evolutionary path we took, but I'm not necessarily sure whether that can be generalised across the universe."

For instance, it is possible that an alien species in a low-oxygen environment could harness heat from geothermal activity, such as volcanoes, to do their smelting, says Waldmann.

Such alternatives aren't portable like wood and fossil fuels, says Balbi, which may mean any civilisations would need to stay close to the heat.

If the oxygen bottleneck is a limiting factor, say Frank and Balbi, it might help solve the Fermi paradox, which questions why there are no signs of advanced intelligent alien life if it is apparently likely, given the size of the universe.

"The Fermi paradox is telling us something about the nature of life in the universe that we don't understand," says Ian Crawford at Birkbeck, University of London. "In that context, all potential solutions to the Fermi Paradox are of interest." ▮

Homework

1. Read only the title of the text. What do you already know about the topic?

2. Does the text seem reliable? Why?

3. Do you think the text will meet your expectations? Why?

4. After reading the text quickly, write down at least 5 keywords. What do they mean?

5. After reading the text more carefully, divide it into sections, e.g. introduction, main part(s), conclusion etc. Give a content-related keyword to each section.

Section	Content-related keyword

6. Write down the main idea of each paragraph using *one or a few words only for each.*

Paragraphs	Main idea
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	

7. Combine your answers to questions 4-6 into a single short summary of one paragraph.

8. Why or how is the article important and what is/are the most important point(s) - the big idea?

9. Write down a reference to the article according to APA referencing guidelines.

Now combine these answers and be prepared to present them in two minutes in class (No PPT needed). *Rehearse your short speech out loud* before coming to class.

Notes (key words only – do not write a script)

Plenary

1. Discuss: plenary: What was the aim of the activity?

2. In groups of 3-4: present your answers in 2'

IV- PROJECT: SOURCE SELECTION

- ARTICLES 1-4: GENERAL INFORMATION vs. CURRENT RESEARCH (News articles vs. current research/popularized scientific articles)

Title	***Source***	***General info***	***Current research***	***Length (min 500 w.)***	***Suitable as current research articles 2-5? Justify***
Google's deepfake hunter sees what you can't—even in videos without faces (link)	Science Daily			721 words	yes - no
AI-generated scientific hypotheses lag human ones when put to the test (link)	Science			915 words	yes - no
The unseen cost of convenience: Study finds EV fast chargers double local air pollution (link)	Natural News			1172 words	yes - no
Brain device that reads inner thoughts aloud inspires strategies to protect mental privacy (link)	Science			957 words	yes - no
Humans keep building robots that are shaped like us – what's the point? (link)	The Conversation			1046 words	yes - no

➢ CURRENT RESEARCH ARTICLE 5: SCIENTIFIC ARTICLE (NOT A REVIEW/SUPERSTUDY)

1. **How do you recognize a scientific article? Look at this example.**
   - Content: ……..
   - Length: ……..
   - Authors: ……
2. **Fill in the following table.**

Title	***Source***	***Scientific article? (tick)***	***If scientific article, research or review (circle)***	***Suitable as article 1?***
Krylov diagonalization of large many-body Hamiltonians on a quantum processor (link)	Nature		Research – Review	yes - no (superstudy)
Artificial intelligence (AI) applications for marketing: A literature-based study (link)	ScienceDirect		Research – Review	yes - no (not really a scientific article)
Still booting after all these years: The people stuck using ancient Windows computers (link)	BBC		Research – Review	yes - no
Cooperative robotic exploration of a planetary skylight surface and lava cave (link)	Science Robotics		Research – Review	yes - no
Deep Speech Synthesis and Its Implications for News Verification: Lessons Learned in the RTVE-UGR Chair (link)	MDPI		Research – Review	yes - no
Will your next surgeon be a robot? Autonomy and AI in robotic surgery (link)	Science Robotics		Research – Review	yes - no

V – PITCHING AN IDEA

*How have you exposed your ideas so far ? Do you know what a **pitch** is?*

The purpose of a(n) (elevator) pitch is to …

Discover what it is exactly in this **video**[iii].

To sum up:

1. A pitch shrinks your idea—values, promise, and character—into one statement. It answers **4 main questions**:

    1) What is the initial **problem**? (e.g. water pollution)
    2) What is your **idea/solution**? What does it do? (e.g., one specific novel cleaning technique)
    3) What is the **value/benefit/impact** of your idea? How does it differ from other similar solutions? Why is it interesting/exciting? What makes it unique? (e.g., explain why this technique is more cost- or time-efficient to tackle water pollution)
    4) What do you want from your audience to do next? (**CTA/summary/conclusion**) (e.g., explain that you are looking for partners to pilot this project)

What are the characteristics of a pitch? It should be:

VI – PITCH STRUCTURING

1. Establish the problem or need
2. Introduce the solution or offering
3. Share the value/benefit or your idea (what makes you different)
4. Optional conclusion/summary

+ Provide evidence or support/example

***SpaceDev* Exercise: identify the pitch structure**

SpaceDev was founded with the vision of producing innovative and affordable space solutions to enable humanity to explore and develop space through practical and profitable

steps. To realize this vision, we are developing evolutionary building blocks – starting with affordable microsatellites and their associated subsystems, and safe affordable propulsion and launch systems. To retain and build the team necessary for these developments, SpaceDev is fostering a corporate culture that attracts and retains creative, practical, and energetic employees who are driven to "make space happen".

VII – AUDIO: JILL HARLAND'S 2016 PITCH[iv]

1. What is her solution? What examples does she provide to explain the technology?

2. Explain the gist of her innovation. What solution does it offer?

3. Why do you think it was a good/bad pitch? (structure, tone, gestures, voice, etc.)

VIII – FIND YOUR PROJECT TOPIC

Personal research

1. Individually, think of **specific**, **innovative** and **recent** research projects and focus on what they offer as a solution to a problem. Browse the internet and look up for **recent research findings** in your field of study. Choosing a **trendy/hot topic** will greatly help you find resources throughout the year.

Now, get up and discuss project ideas with your classmates.

2. Do not directly ask others about their project idea or immediately share yours. Use small talk (why you are personally interested in such or such topic) and find round-about ways to direct the conversation towards the project.

Expose your project topic to your classmates. Give advice about theirs. Avoid any negative comments.

IX – START ON YOUR PROJECT: INSTRUCTIONS

1. Download the Powerpoint template and set up a shared PPT document for your future project.
   Indicate the title of your project and copy and paste the link (click on 'paramètres du lien', '***personnes UMONS***' and share in '***consultation possible***' access) of your shared PPT into the shared Excel sheet *Project: list of topics* (see *Project Exam: info & deadlines* on Moodle) so that all students from your class can access your file without the possibility of modifying it.
2. Determine the goal of the project / the problem addressed by the project (to do so, answer the **four questions** from **'Pitching an idea'**) in one or two paragraphs.
   Your text must be formal. You may struggle with formal academic style, but don't worry. It is a learning process!
   - Please write a continuous text. Do not simply "answer" the questions as if it were a questionnaire (and even less so include them in your text!).
   - Create a structured text (see **'Pitch structuring'**).
3. Prepare yourselves to pitch this idea orally next week under 90 seconds. Find one popular science article to give some details about your project and use it to give one (very quick) example in your pitch. You can keep this article for your project later. **Do not read a text! You are allowed to have key words.**

   Your written project idea must be different in style from your oral pitch, i.e., the former is to inform in a neutral way, the latter is to convince or sell your idea.

X – PRACTISE FOR YOUR PROJECT : PITCH YOUR PROJECT

1. Pitch your project to your classmates (move around in the class). Be convincing and enthusiastic about the project you are defending. Imagine we can only fund one project among all the projects presented today.

2. Take notes about at least 2-3 projects.

3. Roundtable: one student explains the project of another student and give their impressions. Why did you find it interesting? Give some details that stood out.

UNIT 1: DEFINING, DESCRIBING & QUALIFYING

I – SHOW AND TELL

Instructions

Choose an object that you are attached to or that you find particularly interesting/intriguing. Avoid common objects such as smartphones, PS5, etc. Go for something that will not be easy to guess by the rest of the class. You will have to describe this object in class. The other students will be allowed to ask questions if necessary.

In-class

1. In groups of 4/5, take turns to describe the object you have chosen and try to guess what it is by writing your answer on a small piece of paper.

2. Do not give your answer orally during the description.

3. The student who guesses the most objects wins.

II - VOCABULARY – DEFINING & DESCRIBING[v]

Practise on Quizlet

A. Categorization

One of the simplest ways of defining a word is by using the word **"to be"** and assigning it to a general category.

- The definition answers the implicit question: *"What (sort of thing) is X?"*

It is an object • component • instrument • tool • device • apparatus • machine • system • technology • symptom • examination • test

POTENTIAL MODIFIERS

It's a kind of / sort of / type of (tool)

- *Lethargy is a **sort of symptom** that is often linked to emotional trauma.*
- *A colonoscopy is a **kind of examination** used to diagnose intestinal disorders.*

B. Constituent elements

A word can be defined by enumerating and describing its parts or components.

- Implicit question: *"What are the basic elements of X?"*

It is made of • It consists of / in[G. Notes 1] • It is composed of • It comprises • It includes

POTENTIAL MODIFIERS

typically • basically • mostly • mainly

- *The Department **consists basically of** three divisions: gynecological oncology, endocrinology, and urogynecology.*
- ***Typically**, research teams are multidisciplinary, **comprising** clinical, scientific, and computer specialists.*

C. Location

Definitions can be refined by indicating position or direction.

- Implicit question: *"Where is X?"*

It is situated / found / located / placed • It moves / proceeds / progresses (+ preposition)

FREQUENTLY USED PREPOSITIONS

above • along • at the bottom / top • behind • beneath • below • beside • between • in front of • inside • near • next to • in the middle • on the right / left-hand side • opposite • through • towards • underneath • within

- *A stent is a tubular device **placed inside** the artery.*
- *The microscope has an adjustable diaphragm **located beneath** the specimen.*

D. Use

Definitions may explain the use of an object.

• Implicit question: *"What is X for?"*

It is to / for (+ -ing) • It is used to / for (+ -ing)

POTENTIAL MODIFIERS

above all • commonly • essentially • frequently • mainly • typically • widely

- *The money is intended **essentially for financing** the ambulance service.*
- *Heparin **is widely used to** prevent blood clotting.*

E. Shape

We often make definitions by indicating the shape of an object.

• Implicit question: *"What does X look like?"*

It is circular / round / oval / curved • It is square / oblong / triangular / rectangular • It is hollow / solid / rigid / flexible • It is narrow / wide / thin / thick / fat / flat

POTENTIAL MODIFIERS

relatively • quite • fairly • about • roughly • approximately

- *The pancreas is a **fairly narrow**, **flat** organ located deep in the abdominal cavity.*
- *In ultrasonic liposuction, fat is extracted by a **thin**, **hollow** canula.*

N.B. – One simple way of defining a word is by using the structure: **NOUN + "-SHAPED"**.

cube-shaped · cone-shaped · disc-shaped · diamond-shaped · Y-shaped · tree-shaped

➢ *An otoscope is a **cone-shaped** instrument used for examining the ear.*

F. Relative pronouns

In order to gain in precision, definitions often resort to the use of relative pronouns: **who · whose · (of) which · that** …

For people: who · whose

➢ *It is a team **consisting of** 4 doctors **whose** expertise is internationally recognized.*

For objects: which · that

➢ *The nasal septum **is made of** cartilage and bone **which / that** is covered with a lining (mucosa).*

III - VOCABULARY: QUALIFYING[vi]

Practise on Quizlet

1. Adjectives

▮ Importance

important · significant · crucial · meaningful · relevant

➢ *What Bachelard wrote in 1934 is still **relevant** to contemporary scientific problems.*

▮ Dimension

enormous · tremendous · huge · widespread ≠ small-scale · tiny · minute

➢ *Blood plasma contains **minute** quantities of creatinine.*

▮ Hierarchy

major · main · chief · leading · primary · foremost · outstanding ≠ • secondary · minor · common • average · typical · standard

➢ *Little by little the patient's condition **worsened**.*

▮ Intensity

dramatic · striking · acute · steep · sharp

➢ *There has been a **sharp** decline in the quality of river water.*

▮ Aptitude and utility

• appropriate · suitable
• useful · reliable · efficient

➢ *If electromagnetic radiation of **suitable** wavelength falls upon the metal, electrons are ejected.*

2. Adverbs

▮ *Essence*

essentially · fundamentally · basically · on the whole

➢ ***Basically**, there is little difference between the two processes.*

▮ *Manner*

carefully · steadily · thoroughly · accurately · properly

➢ *The 1st law of thermodynamics was not **properly** formulated until the middle of the 19th century.*

▮ *Negation*

hardly (any) · scarcely (any) · barely (any)

➢ *After processing, **hardly any** of the nutrients remain.*

▮ *Degree*

a little · a bit · slightly

relatively · quite G. Notes 5 · rather · fairly

very · extremely · exceedingly

➢ *The results were **slightly** better than last year.*

▮ *Approximation*

nearly · almost · roughly · more or less · virtually · to a certain extent · partly

≠

totally · quite G. Notes 5 · entirely · utterly · thoroughly

➢ *For most people, living without a motor car would be **utterly** unacceptable.*

IV - PRACTISE

1. Describe an object. Look at the following example:



*A mouse **is** a device **which** consists basically of a laser or LED mechanism in a plastic case **and/that** is used to control a cursor with one hand.*

2. Construct sentences with the same pattern, selecting phrases from each box.

A vulcanologist • A rock scientist • A beaker • Goggles • A microscope • A bar code reader	→ IS / ARE	a geologist • a tool for holding liquid • a tool for magnifying small objects • a hand-held or stationary input device • a volcano scientist • a type of safety glasses

↓ WHICH/THAT/WHOSE/WHO

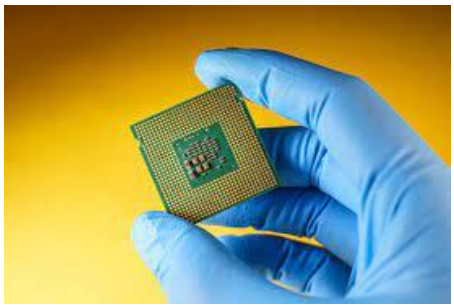
is (mainly used) to/for …• is located in… • who typically works for/in	← AND THAT	is used to capture and read information contained on little vertical lines • job could be for instance to find iron in Earth's bedrock • determines the likelihood of an imminent eruption and studies the past eruptions of a volcano • are designed to protect the eyes from spillings • is typically made of glass • consists of several lenses

3. Describe these objects to someone who cannot see using the 'Useful Vocabulary':

- **What sort of thing is it?**
- **What are the basic elements of X?**
- **Where is it found/used?**
- **What is it used for?**
- **What does it look like (colour, shape)?**













VI - AUDIO – THE HELIX[vii]

A. PREPARATION

Match the words with the definitions.

spiral – to perceive – prevalent – restricted – renowned – a corkscrew

a device for removing corks from bottles .....

a three-dimensional curve around a central point .....

limited .....

famous .....

to notice or realise .....

very common in a particular place .....

B. LISTENING COMPREHENSION

- ***1st listening***

Take notes of the gist.

- ***2nd & 3rd listening***

Are the sentences true or false? Justify your answers orally.

1. The speaker starts the talk with two examples of helices found in nature.	True	False
2. A helix is a three-dimensional object.	True	False
3. The main advantage of a spiral staircase is its strength.	True	False
4. DNA contains the genetic information for most, but not all, living things.	True	False
5. A mirror image of a helix can appear to change direction.	True	False
6. The screw is an example of a conical helix.	True	False
7. Helices are much more common in manufactured objects than in natural objects.	True	False
8. It is a mystery why there are so many helices in the world.	True	False

C. **VOCABULARY**

Write the adjective form of the word in brackets in the gap.

1. To understand this, we will use a complicated …… formula. (mathematics)
2. Doctors are now confident that the condition is caused by a …… defect. (genes)
3. When bamboo is cut, you can see it is a straight pole with a …… hole down the middle. (cylinder)
4. Two-…… space can be viewed as a projection of the physical universe onto a plane. (dimension)
5. The pattern is made up of …… colours: white and black. (alternate)
6. The iris of the eye is composed of …… muscles which you can see as the brown, blue or green colour of the eye. (circle)

D. **DISCUSSION**

Do you ever watch scientific lectures or talks like this online? If yes, on which websites?

UNIT 2: PURPOSE & PROCESS

I - VOCABULARY – PURPOSE & PROCESS[viii]

Practise on Quizlet

1. Purpose

→ Purpose is concerned with questions of **"why"** or for **"what reason"** something is done.

▮ *Nouns*

the purpose • function • use

➢ *Metal is increasingly used in architecture for structural **purposes**.*

the aim • goal • target • objective

➢ *The main **objective** of hydrology is to study the physical and chemical processes of the water cycle.*

▮ *Verbs*

it is designed to / for[G. Notes 23] • is devised to / for • is planned to / for • is aimed to / for • is responsible for

➢ *The new safety measures were **designed to** reduce the risks of contamination.*

➢ *A cooling system was **devised for** the preservation of clinical specimens.*

it functions (as) • operates (as)

it provides • supplies

➢ *The haemoglobin **provides** the extra oxygen.*

▮ *Other expressions*

in order to / so as to • so that

N.B.

- ***"In order to / so as to"** have basically the same meaning as **"to"**. They are merely stronger ways of expressing the idea of purpose.*

- ***"So that"** is followed by a clause composed of subject + verb.*

  ➢ *Platinum contacts should be used **to** prevent oxidisation.*
  ➢ *Platinum contacts should be used **in order to / so as to** prevent oxidisation.*
  ➢ *Platinum should be used **so that oxidisation can be prevented**.*

2. Process

→ Process is concerned with questions of **"how"** or **"in what way"** something is done.

▮ *Adverbial and prepositional phrases (see Unit 5)*

by means of · through · thanks to · via

➢ *The brain was damaged **through** lack of oxygen.*

therefore · thus · thereby

➢ *He switched off the computer, **thereby** losing all the data.*

▮ *Verbs*

to enable · make (it) possible (for) · allow · permit

➢ *It is the carbon compounds that are responsible for the chemical reactions that **enable / make it possible for** the cell to grow.*
➢ *Visibility was poor – the plane was not **allowed / permitted** to land.*

N.B. – Meaning

- ***"To enable / to make possible"** have the primary meaning of making something feasible or physically possible.*
- ***"To allow / to permit"** have the primary meaning of making something possible by giving authorisation or permission.*

*However, in modern English, the meaning of **to allow / to permit** is often extended to physical possibility.*

The Erasmus grant { **enabled** / **made it possible for** / **allowed** / **permitted** } him to go on studying

TO DESIGN VS. TO DEVISE

1. 'The car was _____ in Italy and made in Brazil.'

2. 'The company has _____ a computer programme that allows people to design their own homes.'

→ to _____: to plan and organize the structural elements of an object/project (often on paper)

→ to _____: to elaborate, invent a new system or solution involving careful thought and innovation

Note:

- 'TO' is used before a verb
  e.g. 'These measures are designed TO reduce pollution.'
- 'FOR' is used before a noun (or verbal nouns, often ending in '-ing')
  e.g.: 'Computers were designed FOR processing massive amounts of data.'
  e.g. A new system was devised FOR underwater exploration.'

IN ORDER TO / SO AS TO / SO THAT

1. I regretfully must withdraw from this campaign _____ take care of my wife who was just recently diagnosed with Parkinson's disease."

2. I regretfully must withdraw from this campaign _____ I can take care of my wife who was just recently diagnosed with Parkinson's disease."

→ so that + …

→ in order to/ so as to + …

THEREBY (+ _____)

Students perform in hospitals, thereby _____ (gain/gaining) a deeper awareness of the therapeutic power of music.

II - PRACTISE

1. **Make comments about the use of 2 items (from tne pic below) using phrases such as:**

aims to - purpose - goal - designed for - provide - in order to
via - thereby - enable - by means of - make it possible

Computer parts



2. Here is a list of "Purposes". Imagine at least 2 "Processes" for 3 of them.

- To save money
- To get fit
- To travel safely
- To cook a healthy meal
- To improve writing skills
- To practice speaking in English
- To make music
- To design a game
- To reduce stress
- To sleep better

III – AUDIO: SLEEPING PILOTS AND CHAOS THEORY[ix]

A. BRAINSTORMING

1. Give examples of both major and minor accidents.
2. Were they brought about by faulty technology or human error?
3. Was any action taken as a result of the accident?
4. Give examples of how technology can be used to increase safety.

B. LISTENING

1. Human error accounts for 18% of air accidents. True/False? Justify.
2. How often do pilots generally fall asleep while piloting?
3. What is pilot fatigue due to?
4. What does ENRI stand for?
5. What are the biological functions that ENRI can process?
6. Why is this system particularly efficient?

C. PROBLEM/SOLUTION IDENTIFICATION

In the transcript below, find the excerpts which examine the PROBLEM and the excerpts which indicate SOLUTIONS (why was it created – purpose & How is it achieved – process).

Despite the thousands of parts which are used in the construction of aeroplanes, very few air-crashes are caused by component failure. It is, in fact, human error that is responsible for 80% of air accidents. This explains why one of the primary objectives of the aircraft industry today is to devise/design ways of improving these figures. One area of particular concern is pilot fatigue. This problem is far more widespread than is commonly realised and Dr. Mark Rosekind

claims that "70% or more pilots say that they have fallen asleep at least once while piloting". Fatigue is particularly linked to the crossing of time zones which can severely disturb the circadian rhythm.

The Japanese company, Electronic Navigation Research Institute (ENRI), is currently in the process of developing a monitoring system which will be able to provide early warning of fatigue by means of voice recognition techniques.

Biological functions, such as blood pressure, heartbeat and voice, have a fractal structure which can be identified. Under stress or fatigue, these physiological manifestations undergo change and consequently, as can be seen in the figure, the pattern alters, gradually losing its structure and becoming chaotic. By using the mathematics of chaos theory, Kakuichi Shiomi, chief researcher of ENRI, has been able to design/devise software which enables these minute distinctions to be identified so that the pilot's voice pattern can be matched with a control model of an alert voice. It takes no more than 10 seconds for data samples to be processed which means that fatigue can be measured in real time without interfering with the normal activity of the flight crew.

According to Shiomi, the system makes it possible to detect tiredness in test subjects 10 to 20 minutes before they become aware of it themselves.

IV- PRACTISE FOR YOUR PROJECT: PRESENTING OBJECTS/ACTIVITIES

Instructions

Prepare a PowerPoint presentation about 1-2 key objects/activities from your project. Your task will be to:

- define/describe and qualify these key objects/activities from your project.
    - What sort of object is it?
    - What are the basic elements of this object?
    - Where is it?
    - What does it look like (colour, shape)?
- explain how these objects were created/activities were born (process, i.e. origin, design, conception) and why these objects/activities are important (purpose).
    - What are they used for?
    - How were they designed?
    - How are they used?
    - Were they designed specifically for certain tasks? Are they common?

To do so, you will need to use the vocabulary from 'Defining, Describing & Qualifying' and from 'Purpose & Process'.

You need to:

- Add information on your project PowerPoint. Use pictures only, no words and put each object on a separate slide if you have more than one.
- Present your objects/activities orally to other students in small groups. Do not read a text. Use a few notes only (keywords on a small piece of paper).

In-class :

2. Presentations in small groups.

▪ Presenters: Present your objects to six students. Do not read a text. Use a few notes only (keywords on a small piece of paper) and a PPT slide.

▪ Listeners: Take notes of the information in the table below.

▪ Be ready to report back on other students' objects.

Object 1:	Object 2:
Object 3:	Object 4:
Object 5:	Object 6:

3. Report on your classmates' objects.

UNIT 3: MEASURING & QUANTIFYING

I - VOCABULARY (1): MEASURING & QUANTIFYING[x]

Practise on Quizlet

N.B. – Many of the words in this section have multiple grammatical forms, e.g.

NOUN : ➢ *a blood **sample***
VERB : ➢ *the bacteria was randomly **sampled***

1. Adjectives

deep ≠ shallow · far ≠ near · fast ≠ slow · heavy ≠ light · high ≠ low · long ≠ short · odd ≠ even · thick ≠ thin · wide / broad ≠ narrow

➢ *All prime numbers are **odd** numbers.*

accurate ≠ inaccurate · average / mean · standard ≠ sub-standard

➢ *The **mean** density of Mercury is similar to that of the Earth.*

2. Nouns

amount · extent · measurement · range · size · span · speed

accuracy · average · level · mean · rate · scale · stage · step

➢ *The **rate** of acceleration is expressed in metres per second per second.*

check · study · survey

area · circumference · cross-section · diameter · radius

➢ *The **cross-section** of the wire is 0.22 mm^2. (nought point two two square millimetres)*

▮ *Rules for noun formation – suffixes*

*ADJ / VERB + **-th / -t** (+ VOWEL CHANGE)*

dep**th** · hei**ght** · long / leng**th** · wei**ght** · wid**th**

*ADJ + **-ness***

hard**ness** · heavy / heavi**ness** · near**ness** · thick**ness**

*VERB + **-ment***

to develop / develop**ment** · measure**ment** · move**ment**

3. Verbs

▮ *Rules for forming verbs*

NOUN/ADJ + Ø (NO CHANGE)

to narrow ≠ to thin · to range / to span / to extend[6. Notes 1] / to reach · to rate / to check / to monitor · to record / to plot

➤ *The trajectory of the missile was **plotted** on a graph.*

*NOUN/ADJ + **-en***

to deep**en** · to length**en** · to short**en** · to thick**en** · to wid**en**

➤ *The river **widens** when it leaves the canyon.*

*NOUN/ADJ + **adv particle***

to check **up** · to level **off** · to slow **down** ≠ to speed **up** · to step **up** · to work **out**

➤ *The speed of the neutrons is **slowed down** by the beryllium moderator.*

4. Structures

Dimensions can be expressed by 4 different structures.

It **is** 56 m	**high** / wide / long / thick	
It **is** 56 m	**in height** / in width / in depth / in diameter	
Its	radius **is** 5 m / length / cross-section / circumference	
It has	**a** diameter **of** 10 m / a length of / a weight of	



5. Other measurements

Area

To obtain the area, you **multiply** the length **by** the width.
It measures 10 cm **by** 10 cm. The area is 100 cm^2 (a hundred **square** cm).
πr^2 (**pi r squared**) G. Notes 2 • $\sqrt{x}$ (the **square root** of x)

Volume

The volume is 1,000 cm^3 (a thousand **cubic centimetres**).
x^3 (x **cubed**) • $\sqrt[3]{y}$ (the **cube root** of y)

Power

x^9 (x **to the power** nine / x **to the** ninth)
x^{-9} (x to the **power minus** nine / x to the **minus ninth**)

6. Approximate measurements



These can be expressed
by means of **adverbial modifiers**.

It is	**approximately** 5 cm long
	about / roughly / more or less
	almost / nearly
	a little over / slightly under

7. Questions

Note the question forms.

It weighs 10 kg	→	**How heavy** is it? / **How much** does it **weigh**?
		What does it **weigh**?
It is 5 km away	→	**How far** (away) is it? / **How many** kilometres **away** is it?
		What is the **distance**?

CUSTOMARY & METRIC UNITS

CUSTOMARY UNITS

LENGTH

1 foot (ft) = 12 inches (in.)
1 yard (yd) = 3 feet
1 yard = 36 inches
1 mile (mi) = 1 760 yards
1 mile = 5 280 feet

CAPACITY

1 pint (pt) = 2 cups
1 quart (qt) = 2 pints
1 quart = 4 cups
1 gallon (gal) = 4 quarts

WEIGHT

1 pound (lb) = 16 ounces (oz)
1 ton (T) = 2 000 pounds

TEMPERATURE

water freezes = 32°F
water boils = 212°F
normal body temp. = 98.6°F

METRIC UNITS

LENGTH

1 centimeter (cm) = 10 millimeters (mm)
1 decimeter (dm) = 10 centimeters
1 meter (m) = 10 decimeters
1 kilometer (km) = 1 000 meters

CAPACITY

1 liter (L) = 1 000 milliliters (mL)
10 deciliters (dL) = 1 liter

WEIGHT

1 gram (g) = 1 000 milligrams (mg)
1 kilogram (kg) = 1 000 grams

TEMPERATURE

water freezes = 0°C
water boils = 100°C
normal body temp. = 36.6°C

CAPACITY

1 L ≈ 1.06 qt
1 gal ≈ 3.8 L

LENGTH

1 in. = 2.54 cm
1 m ≈ 39.37 in.
1 m ≈ 1.09 yd
1 km ≈ 0.6 mi
1 mi ≈ 1.6 km

WEIGHT

1 oz ≈ 28 g
1 kg ≈ 2.2 lb

$°F = °C \times 1.8 + 32$	$°C = (°F - 32) \div 1.8$
Convert Celsius to Fahrenheit	*Convert Fahrenheit to Celsius*
© CalculatorSoup	© CalculatorSoup

1 foot (ft)= 30 cm
1 yard (yd) =0.91 m
1 pint (pt) = 47.3 cl
1 pound (lb)= 454 g
1 gallon (gal, US) = 3.78 l
1 gallon (gal, UK) = 4.5 l
1 stone (st)= 6.3 kg

www.unitconverters.net

II - PRACTISE: KEYPOINTS 1-4

1. In pairs or in groups of 3, prepare a mind map about measurement (depth, weight, height, width, length, speed, imperial system units). Do <u>NOT</u> look at the 'Useful Vocabulary' section.

e.g.





2. Answer the following questions.

a. What can be odd or even?

b. How many kilometres is a mile? 1 mile = ……. km

c. To span = …….. (synonym)

e.g. A series of bridges span the river.

d. To speed up >< ……..

e.g. They have speeded up production of the new car.

3. Write sentences to describe the following landmarks. [1 ft = 0.3m ; 1 lb = 453.6g]

- **Big Ben**



Height: 97.5m

Length: 12m

Width: 12m

Weight (bell): 13.7 tonnes

Weight (pendulum): 310 kg

Length (pendulum): 4.4m

Depth (below ground): 4m

……

……

……

……

……

……

……

……

……

- **Queensferry Crossing**



River: Firth of Forth

Length: 1.7 miles (2.7 km)

Height: 696 ft (212 m)

Weight (bridge deck): 35,000 tonnes

……

……

……

……

……

……

……

……

……

4. Look at the pictures and write questions with:

- **how + adjective** *e.g. How big…?*
- **what** *e.g. What is the height…?*



Picture 1



Picture 2



Picture 3



Picture 4

Picture 1:

……

……

Picture 2:

……

……

Picture 3:

……

……

Picture 4:

……

……

III - VOCABULARY (2): NUMBERS AND BASIC MATHEMATICAL SIGNS[xi]

- **0**

The following are different ways of saying **0** in English for various contexts:

OH	After a decimal point	5.03	five point oh three
	In telephone numbers	67 01 75	six seven oh one seven five
	In bus numbers	N° 703	get the seven oh three
	In hotel room numbers	Room 206	room two oh six
	In years	1904	nineteen oh four
NOUGHT	Before a decimal point	0.03	nought point oh three
ZERO	For the integer	0	the number zero
	For temperature	-8°C	eight below zero
NIL	In football scores	5-0	Belgium beat England five nil
LOVE	In tennis	15-0	fifteen love to Federer

- **LARGE NUMBERS (*British English*)**

108	a/one hundred **and** eight
336	three hundred **and** thirty-six
1,074	one thousand **and** seventy-four
2,469	two thousand four hundred **and** sixty-nine
250,000	250,000 a quarter of a million *or* two hundred **and** fifty thousand
331,749	three hundred **and** thirty-one thousand seven hundred **and** forty-nine
3,146,658	three million one hundred **and** forty-six thousand six hundred **and** fifty-eight
1,000,000	one million
1,000,000,000	one billion
1,000,000,000,000	one trillion
1,000,000,000,000,000	one quadrillion

- **BASIC ARITHMETIC**

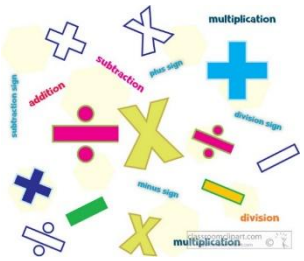
4 + 4 = 8	four ***plus*** four ***equals*** eight	(verb: to add)
9 – 2 = 7	nine ***minus*** two equals seven	(verb: to subtract/deduct)
5 x 5 = 25	five ***times*** five equals twenty-five *or* five ***multiplied by*** five equals twenty-five	
8 ÷ 2 = 4	eight ***divided by*** two equals four	
2^2	two squared	
2^3	two cubed	
2^4	two to the power (of) four *or* two to the fourth	
√ x	the square root of x	
$\sqrt[3]{x}$	the cube root of x	
π	pi	
1/1024	1 over 1024	
(x – y) / z	x minus y, over z	
x – y / z	x, minus y over z	
10 l/min	ten litres ***per*** minute	
0.6-63mg	nought point six ***to*** sixty-three milligrams	
5/8	five eighths	
4/9	four ninths	

- **DECIMAL FRACTIONS**

0.01	nought point oh one
0.001	nought point oh oh one
1.1	one point one
1.23	one point two three
21.57	twenty-one point five seven
2.6666666…	two point six recurring
2.5 million	two point five million

➢ **ORDINAL NUMBERS**

1st	first	4th	fourth	11th	eleventh
2nd	second	5th	fi**fth**	12th	twel**fth**
3rd	third	6th	sixth	13th	thirteenth
		7th	seventh	14th	fourteenth
		8th	eigh**th**	15th	fifteenth
		9th	ni**nth**	…	
		10th	tenth		
20th	twent**ieth**	100th	hundredth		
21st	twenty-first	1000th	thousandth		
22nd	twenty-second	1,000,000th	millionth		
23rd	twenty-third				
24th	twenty-fourth				
25th	twenty-fifth				
…					





Useful website: http://www.englishnumber.com/

IV - PRACTISE: NUMBERS AND MATHEMATICAL SIGNS[xii]

1. Read the following calculations out loud.

$9^2 = 81$

$5^3 = 125$

$10\ km^2 = 1\ km \times 10\ km$

$\sqrt[3]{216} = 6$

$3^4 = 81$

$4^{-5} = \frac{1}{1024}$

2. Read the following out loud, paying particular attention to stress patterns:

1 km = 0.621 mile	⅜s of a pound	2.718281828	$50.25
0.00000001	1 cm = 0.39 inches	45 mins = ¾ hr	1 kg = 2.2 pounds
€25,000.21	20 mins = ⅓ hr	3.14159	1 m = 1.09 yards
9.11 secs	1 g = 0.035 ounces	19.07%	53.6%

3. Listening comprehension: Take notes of the numbers and mathematical signs that you hear.[1]

I.	a.	II.	a.	III.
	b.		b.	
	c.		c.	
	d.			
	e.			

[1] *Source: Unit 2: ex. 9*

4. Listening comprehension: Fill in the correct numbers (in digits) for each statement relating to a scientific experiment or finding.[2]

1. As this was a dosage of _____ milligrams of fluoride per kilogram body weight, the equivalent amount needed to achieve a similar peak in a _____ -kilogram child would be _____ milligrams of fluoride.

2. The sensitivity of the assay was _____ picomoles.

3. The output impedance is about _____ ohms at the _____ volt end and _____ ohm at the _____ volt end of the range.

4. _____ -amp _____ -core mains flex is used for the mains input which connects straight to the PCB.

5. Inserting a few atoms of potassium makes the compound a superconductor which, below a critical temperature of about _____ kelvins, conducts electricity with no resistance.

6. This shows that where _____ moles of hydrogen gas combine with _____ mole of oxygen gas to form _____ moles of liquid water, at a pressure of _____ atmosphere, and a temperature of _____ kelvins, the enthalpy change is _____ kilojoules.

7. Isolated young mice squeak repeatedly at frequencies of _____ kilohertz to _____ kilohertz, until their mother comes and returns them to the nest.

8. In a similar form of these experiments, conventional, _____ -nanosecond laser pulses were used.

[2] *Source: Unit 2: ex. 10*

V– VIDEO: HOW MANY GIGABYTES CAN A HUMAN BRAIN STORE? [xiii]

According to you, how much information the human brain might store?

A. LISTENING COMPREHENSION

- ***1st listening***

Take notes of the gist. Pay a particular attention to the numbers that are given in the video.

- **2nd & 3rd listenings**

Formulate the questions correctly and then answer.

1. What do we already know about the human brain?

- Weight?

- Composition?

- Function?

- Number of neurons?

- Number of neurons for long-term storage?

- Potential storage capacity?

2. What is the difference between storage and retrieval in the brain?

3. Are there any exceptional examples of human memory mentioned?

4. How much data can a single long-term neuron store if we assume 1 byte per neuron?

5. How does the brain actually achieve much greater storage than 1 gigabyte?

VI - REVISION

- Flashcards: Revision flashcards

Take notes of the word you didn't guess (easily):

VII - PRACTISE FOR YOUR PROJECT: POWERPOINT PRESENTATION ABOUT MEASURABLE OBJECTS

Instructions

1. Search the web for basic data/measurements linked to your project. You can either expand on the description(s) given in the first presentation (key "objects" in your project) and further describe them by giving some measurements/key data linked to them, or you can present other measurable items from your project if you prefer. You must present a minimum of **ONE object**.
2. Prepare a short presentation (3-5') including location, **dimensions**, short history and bibliography of each item. Stick to **simple items** (objects, devices, or basically anything that is **measurable** in your project). **Do not use complex graphs with statistics, rates, proportions, etc**. We will work on this later.
3. Make sure to **present these data meaningfully**, that is to say, put them in perspective (remember Purpose & Process - facts are not interesting if we don't understand why they are, in a specific context)
4. Put one picture per slide and describe it. **Do not write the measurements in your PowerPoint** (give them orally). Do not write sentences on your slides. Use illustrations, keywords and bullet points. Use English sources!

In class

1. Presenter: Present your measurable object(s) to six students. Do not read a text. Use a few notes only (keywords on a small piece of paper) and a PPT slide.
2. Listeners: Take notes of the information in the table below.
3. Be ready to report back on other students' objects.

Object 1:	Object 2:
Object 3:	Object 4:
Object 5:	Object 6:

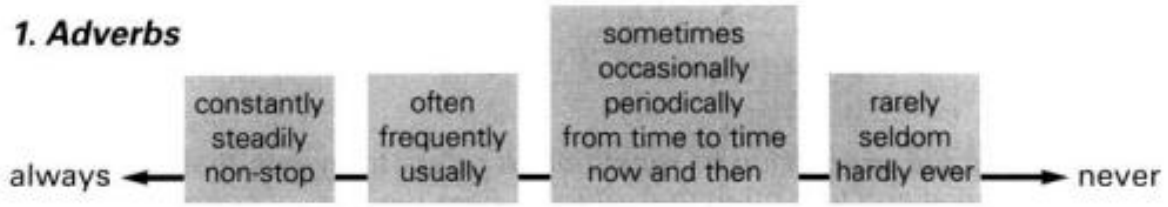
4 Discussion: Feedback about the presentations.

UNIT 4: FREQUENCY

I -VOCABULARY - FREQUENCY[xiv]

Practise on Quizlet

1. Adverbs



- *Politicians **seldom** admit their mistakes.*
- ***From time to time**, there are violent cyclones in the Gulf of Mexico.*

2. Adjectives

- regular · steady · even · constant · non-stop
- irregular · random · unpredictable
- periodic · intermittent · recurrent · repeated · cyclic
- oscillating · alternating · fluctuating

- *There were **recurrent** interruptions throughout the whole meeting.*
- *There is a relatively **even** distribution of the population in Holland.*

3. Nouns

- an oscillation · wave · wave-length · pulse · beat
- a group · set · cluster · range · array · pattern

- *A **series** of remarkable events occurred.*
- *The Galápagos consist of a **cluster** of islands in the Pacific.*

4. Verbs

- to recur · repeat · reduplicate · echo
- to oscillate · fluctuate · vibrate · alternate · beat · swing

- *In politics, public opinion periodically **swings** from left to right.*
- *If the symptoms **recur**, you should consult a doctor at once.*

5. Word formation

NOUN (TIME UNIT) + -ly

hour**ly** · dai**ly**
week**ly** · year**ly**

- *A **weekly** newspaper.*

re- + VERB

to **re**new · **re**build
replace · **re**arrange
reorganise · **re**produce

- *They **rearranged** the furniture in the office*

6. Fixed expressions are also used to indicate:

▮ *Recurrence*

every (month / two weeks / three weeks / other / second / third day) once / twice / three times (**a / per**) minute / day / week

- ➢ *They are producing cars at a rate of 100* ***a / per day****.*
- ➢ *He goes to the hospital for a check-up* ***every other month****.*

▮ *Ratio*

- ➢ ***Two out of three*** *road accidents are caused by alcohol.*
- ➢ *In Denmark,* ***three in ten*** *people speak two foreign languages.*
- ➢ *The normal ratio of girls to boys is* ***100:106****.* (a hundred **to a** hundred and six)

II – GRAMMAR – POSITION OF FREQUENCY ADVERBS

❖ **KEY POINT 1**

1. Place the following adverbs on the frequency line.

always – never – frequently - now and then - hardly ever - constantly



2. Find synonyms for each adverb.

- constantly =
- frequently =
- hardly ever =
- now and then =

3. Rules:

- **IN SIMPLE TENSES (e.g. simple present, simple past…)**

  - 'BE': after 'be'
    e.g. I am hardly ever late.
    Are you often late?

  - OTHER VERBS: between the subject and the verb
    e.g. I never arrive late.

- **IN COMPOUND TENSES (e.g. present perfect, will-future…)**

  before the second verb (NOT the auxiliary verb).

  e.g. He does not often go skiing.
  He doesn't often go skiing.
  Do you always remember to do your homework?
  I have never been to Australia.
  Have you ever been to Australia?



Exception : 'Now and then': before the subject or after the object
e.g. Now and then, she goes to the cinema.
She goes to the cinema now and then.

4. Use a cross (X) to indicate the right position of the frequency adverbs in those sentences.

a. Are you late? (frequently)
b. I arrive late. (never)
c. Even simple chains are quite difficult. (often)
d. The evidence has been contradictory. (often)
e. Temperatures drop below freezing at night in winter. (frequently)
f. Do science experiments provide the data needed to definitively answer a question?(always)
g. Statistics must be used with care. (always)
h. It doesn't work. (often)
i. I have been to Australia. (never)
j. Have you been to Australia? (ever)
k. We need a break. (now and then)

5. Walk around in the classroom and ask your peers the following questions. Make full sentences.

a. How often do you exercise?

b. How often do you go abroad?

c. How often do you work for your English class?

d. How often do you watch programmes in English on TV or on your computer?

e. How often do you read scientific news on the internet?

f. How often do you use your smartphone?

g. What's your best or worst habit?

h. Tell us about a habit you'd like to start having and why.

i. Explain a bad habit you had before university and that you managed to change.

j. Your turn! Write 3 more questions and ask around.

k. Your turn! Write 3 more questions and ask around.

III - VIDEO: HOW STATISTICS CAN BE MISLEADING - MARK LIDDELL[xv]

- ***Warm-up:***

- What do you think "statistics can be misleading" means?
- Why do you think understanding the subtleties of data interpretation matters, especially in science or daily decisions?

Voc: *to lurk = to hide*

- ***1st, 2nd and 3rd viewings – Answer the following questions***

What is the Simpson's Paradox? Explain it with your own words using the three examples provided in the video.

- ***Critical thinking***

1. How can visualizing data (tables, bar charts, scatter plots) help us detect situations where Simpson's Paradox might be hiding?
2. How could Simpson's Paradox affect fairness in machine learning models (e.g., predicting loan approvals, hiring, or medical diagnoses)?
3. Imagine you are analyzing the effectiveness of a new drug. Besides the overall success rate, which other factors (or subgroups) should you consider? Why?

IV – AUDIO: AEROBIC RUNNING[xvi]

1. What's aerobic running ?

2. What shouldn't accumulate in the muscles during this exercise?

3. Give figures related to the heart.

4. How often should you train?

5. How can you avoid monotony during your training?

6. When should you carry out AT?

7. What are hills sessions?

8. How can you relax your feet?

VI - PRACTISE FOR YOUR PROJECT: PRESENTING MORE SPECIFIC DATA

A few weeks ago, you presented basic data (measurements & quantifications) linked to your project. You should now only focus on more specific data. This involves mainly statistics, proportions, percentages, ratios and rates.

Instructions

1. Prepare to describe minimum one graph from your general information articles orally using the 'Vocabulary' from the units 'Frequency' and 'Qualifying'. You may also refer to 'Appendix 4: Describing a graph' and Present Your Data Like a Pro to understand better how to present data on a slide

   Tips:

   - Make sure to present these figures meaningfully, that is to say, put them in perspective (remember 'Purpose & Process' - facts are not interesting if we don't understand why they are in a specific context)
   - Avoid putting too many figures on each slide. One graph = one slide.
   - Do not use sentences on your slides and do not read a text.
   - Use illustrations, keywords and bullet points only.

2. Use your shared project Powerpoint.

In class

1. Presenter: Present your graph to four different students. Do not read a text. Use a few notes only (keywords on a small piece of paper) and a PPT slide.
2. Listeners: Take notes of the key graph information in the table below. Fill in the feedback grid as well (1 per student).
3. Be ready to report back on other students' objects.

Graph 1:	Graph 2:

Graph 3:	Graph 4:

Student's name:

CRITERIA	DESCRIPTION	SCORE
1. Respect of instructions	Fully respects all task requirements (one graph per slide, no sentences on slides, oral description with keywords, meaningful context provided)	4
	Mostly respects the task, minor omissions	3
	Partially respects the task; some instructions not followed	2
	Task not respected (e.g., too many graphs, text-heavy slides, off-topic)	1-0
2. Ideas & content	Very clear, relevant, and meaningful description of the graph; puts figures into context	4
	Clear and mostly relevant ideas; some attempts to give context	3
	Basic description, with little or unclear context	2
	Limited, vague, or irrelevant content	1-0
3. Organization & delivery	Clear and logical oral presentation; smooth flow	4
	Mostly organized and coherent	3
	Some organization, but presentation sometimes unclear	2
	Poorly organized or confusing delivery	1-0
4. Language (grammar & vocabularyç	Accurate language, rich and precise use of vocabulary (esp. frequency & qualifying expressions)	4
	Mostly correct language; adequate vocabulary use	3
	Some errors or repetitive/simple vocabulary	2
	Frequent errors or very limited vocabulary	1-0
5. Visual support	Effective and clear slides: graph is readable, keywords/illustrations support the oral explanation well	4
	Mostly clear and supportive slides	3
	Slides somewhat busy or unclear	2
	Slides difficult to follow, not aligned with oral presentation	1-0
TOTAL		/20

Student's name:

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5. Visual support	Effective and clear slides: graph is readable, keywords/illustrations support the oral explanation well	4
	Mostly clear and supportive slides	3
	Slides somewhat busy or unclear	2
	Slides difficult to follow, not aligned with oral presentation	1-0
TOTAL		/20

4. Discussion: Feedback about the presentations.

UNIT 5: LINKING YOUR IDEAS

I - STARTER

1. Why is it important to link your ideas?

2. How can you show a link between your ideas?

II - VOCABULARY: LINKING YOUR IDEAS (GENERAL)[xvii]

Practise on Quizlet

1. Giving additional information

and • not only / but • as well as •
in addition • furthermore • moreover • besides • apart from

➢ *The flat is too small and* ***furthermore****, it is not in the town centre.*

2. Contrast

but • however • nevertheless • yet[G. Notes 7] • although • (even) though •
in spite of / despite • whereas • on (the) one hand / on (the) other hand

➢ *Geologists are using radio-active analysis of rocks more and more,* ***nevertheless*** *it is not suitable in every case.*

3. Contrast to what people think

actually[G. Notes 8] • in fact •
in actual fact • in reality

➢ *Most people think that Fleming was the father of penicillin, but* ***actually*** *most of the work was done by Florey.*

4. Introducing a cause / consequence

- because • as • since
- consequently • therefore • thus • hence • as a result • thereby[G. Notes 11]

➢ *The arteries become gradually blocked,* ***thereby*** *reduc****ing*** *the oxygen supply to the brain.*

5. Introducing something obvious / generally accepted

obviously • naturally • of course •
clearly • doubtless[G. Notes 9]

➢ ***Obviously****, unless the birth rate decreases, there will be a major food crisis.*

6. Making a general statement

in general • generally speaking •
on the whole • as a rule

➢ ***As a rule****, metals are inflammable.*

7. Clarification

for example • for instance • e.g.[G. Notes 10] •
that is to say • in other words • i.e.[G. Notes 10] • such as • namely

➢ *The quantity of heat is expressed in the same units as energy and work,* ***namely*** *joules.*

III - VOCABULARY: LINKING YOUR IDEAS (CAUSE & CONSEQUENCE)[xviii]

Practise on Quizlet

1. Adverbs and conjunctions

▮ *Cause*

because of • owing to • due to • on account of • thanks to

➢ ***Owing to** his bad health, the minister was forced to resign.*

because • as • since

➢ ***As** he was ill, the minister was forced to resign.*

▮ *Consequence*

consequently • therefore • thus • hence • as a result • thereby G. Notes 11

➢ *The minister was in bad health, **therefore** he was forced to resign.*
➢ *The minister resigned, **thereby** putt**ing** the government in difficulty.*

2. Verbs

▮ *Cause*

to cause • result in • lead to • be responsible for • bring about • give rise to • trigger (off) • spark (off) G. Notes 17

➢ *Illness **led to** the minister's resignation.*

▮ *Consequence*

to come from • result from • arise from • stem from

➢ *The minister's resignation **stemmed from** his bad health.*

3. Nouns

▮ *Cause*

cause • reason • origin • source

➢ *The real **cause** of his resignation was bad health.*

▮ *Consequence*

- consequence • result • outcome • effect
- by-product • spin-off

➢ *The final **outcome** of the illness was the minister's resignation.*

IV - VOCABULARY: LINK WORDS IN CONTEXT

Link words (also called connectors, discourse markers or conjunctions) are cohesive devices, used mainly to relate one sentence to another or to signal the speaker's attitude or style. They can be used in conversation to indicate agreement, contradiction, surprise, etc. Intonation is important as it helps to convey meaning.

1. SIMILARITY

AND

- *I study and I work.*

NOT ONLY/BUT ALSO

- *Not only do I study, but I also work.*
- *He is not only funny, but also he is intelligent.*

AS WELL AS

- *Please proofread for spelling mistakes as well as grammatical errors.*

IN ADDITION

- *In addition, they can be taught to feed themselves if you allow them access to your pantry.*

FURTHERMORE

- *Furthermore, they make wonderful walking buddies.*

MOREOVER

- *Moreover, they'll show you the way home when you're drunk.*

BESIDES

- *Besides not eating much, they won't ever chew on your electric cords.*
- *I'm not ready to get married yet. Besides, I enjoy living alone.*

APART FROM

- *Apart from their house in London, they also have a villa in Spain.*

2. CONTRAST

BUT

- *His mother won't be there, but his father might.*

HOWEVER

- *This restaurant has the best kitchen in town. However, their staff are quite rude.*

NEVERTHELESS

- *I was in so much pain I didn't want to get up in the morning. Nevertheless, I went to football practice as usual.*

YET

- *I've asked you not to leave your dirty socks on the floor. Yet, you keep doing it.*

ALTHOUGH

- *Although the sun was shining, it wasn't very warm.*

(EVEN) THOUGH

- *Anne was fond of Tim, though he often annoyed her.*

IN SPITE OF THIS/DESPITE

- *In spite of his age, he still leads an active life.*
- *In spite of feeling tired we decided to go out.*
- *Despite his age, he leads an active life.*
- *Despite missing the train, he arrived on time for his English lesson.*

INSTEAD OF

- *Tickets will cost only £5, instead of the usual £6.50.*
- *Can't we deal with this now instead of waiting until tomorrow?*

WHEREAS

- *Some of the studies show positive results, whereas others do not.*

ON THE ONE HAND/ON THE OTHER HAND

- *On the one hand, England has the best language schools. On the other hand, it has the worst weather.*

3. CONTRAST TO WHAT PEOPLE THINK

ACTUALLY

- *Most people think that Fleming was the father of penicillin, but actually most of the work was done by Florey.*

IN FACT

- *I used to live in France; in fact, not far from where you're going.*

IN ACTUAL FACT

- *I thought the work would be difficult. In actual fact, it's very easy.*

IN REALITY

- *In reality, human beings and dinosaurs never lived on earth at the same time.*

4. INTRODUCING A CAUSE/CONSEQUENCE

BECAUSE

- *I did it because he told me to.*

AS

- *As you were out, I left a message.*

SINCE

- *We thought that, since we were in the area, we'd stop by and see them.*

CONSEQUENTLY / AS A CONSEQUENCE

- *Many parents today do not have time to cook healthy meals for their children. Consequently/As a consequence, many children grow up eating too much junk food.*

THEREFORE

- *We're going to experience some meteor showers in the next few days. Therefore, the number of miraculous self-healings will rise.*

THUS

- *You didn't tell me you wanted to come. Thus, we won't be taking you with us.*

HENCE

- *We suspect they are trying to hide something, hence the need for an independent inquiry.*

AS A RESULT

- *I've done a pranic healing course. As a result, I've been able to cure my neighbour's sick cat.*

THEREBY (+ -ing verb)

- *The arteries became gradually blocked, thereby reducing the oxygen supply.*

5. INTRODUCING SOMETHING OBVIOUS / GENERALLY ACCEPTED

OBVIOUSLY

- *Obviously, unless the birth rate decreases, there will be a major food crisis.*

NATURALLY

- *Naturally, I get upset when things go wrong.*

OF COURSE

- *Of course, there are other ways of doing this.*

CLEARLY

- *Clearly, this will cost a lot more than we realised.*

DOUBTLESS

- *You will doubtless find a solution.*

6. MAKING A GENERAL STATEMENT

IN GENERAL

- *In general, Japanese cars are very reliable and breakdowns are rare.*

GENERALLY SPEAKING

- *Generally speaking, the more you pay, the more you get.*

ON THE WHOLE

- *On the whole, I'm in favour of the idea.*

AS A (GENERAL) RULE

- *As a general rule, vegetable oils are better for you than animal fats.*

7. CLARIFICATION

FOR EXAMPLE / FOR INSTANCE

- *Wikis vary in how open they are. Some wikis, for example/for instance, allow anybody to edit content, while others only allow registered users to do this.*

E.G.

- *The website has a variety of interactive exercises (e.g. matching games, crosswords and quizzes).*

THAT IS TO SAY

- *Three days from now, that is to say on Friday.*

IN OTHER WORDS

- *They asked him to leave - in other words he was fired.*

I.E.

- *Service charge is included in all prices; i.e. you don't have to leave a tip.*

SUCH AS

- *Web 2.0 technologies, such as wikis, blogs and social networking sites, have changed the way that people use the Internet.*

NAMELY

- *The quantity of heat is expressed in the same units as energy and work, namely joules.*

V - VOCABULARY: LINK WORDS & FORMALITY [xix]

Link words tend to vary according to the formality of the context.

Meaning	*SCIENTIFIC PRESENTATIONS (ORAL)*		*ACADEMIC WRITING*
	Basic form & formal use	***More formal***	***Most formal***
Addition	also first, second etc. … too.	additionally besides further furthermore in addition next last but not least	equally important moreover similarly
Cause-effect	then therefore	as a result for this reason thus	as a consequence consequently hence
Comparison	also like too	as well as both… and… compared to in the same way as likewise neither… nor…	by comparison in common with similarly
Contrast	however	neither…nor… instead nevertheless	conversely in contrast to in opposition to on the contrary otherwise still …whereas…
Time	after a while after that also at last currently earlier eventually finally first, second, etc… in the future in the past last next now	afterward at the same time formerly immediately in the meantime later	concurrently previously simultaneously subsequently
Example	for example	for instance in other words	as an example as an illustration to exemplify
Summary - conclusion	finally therefore	after all all in all at last briefly last on the whole thus to sum up in conclusion	accordingly as a consequence in brief in closing in short in summary to conclude to summarise

VI - PRACTISE

1. Identify what helps to link ideas in the text below[xx] (part of the transcript of a presentation)

"Good morning, everyone. I'm going to talk to you this morning about public health, and in particular, two very different public health issues that are affecting the health of the world today. I'm going to talk first about the topic of obesity, and I will outline some of the major issues related to this current phenomenon. Then, I will go on and do the same for climate change and its impact on health. I'll also give you a summary of the points we have covered.

But before we begin, I have a little story to tell you. Before becoming a lecturer, I worked for Médecins du Monde. For those of you that haven't heard of it, it's an international humanitarian organization that provides medical care both in France and in the developing world. I was based in Haiti, and it was a fantastic experience, at least from a personal point of view. I was working with children and young mothers, and the team was just great. But as you can imagine, the situation was far from ideal: medical supplies were low, and the conditions were just appalling. Consequently, there was a lot of frustration amongst the medical staff, as well as for the Haitians themselves. One of the biggest problems was corruption, the dishonesty of some of those in power. Of course, the fact that there wasn't any real public health system in place didn't help. It meant that supplies often didn't get through, or they were hijacked on the way. And the ones that actually suffered the most were the kids... Of course, the point of that story is to illustrate the fact that the role of public health authorities in ensuring the health of a nation is crucial. So - to get back to the main part of my lecture..."

2. Rewrite the sentences using the transitions in brackets. You may need to modify the structure of the sentence.[xxi]

a. She didn't give any chocolates to her friends. She ate them all herself. (instead of)

b. She was sick and didn't study for a month. She got a good mark in the test. (however)

c. People couldn't reach the mountain by road. It was snowing. (because of)

d. I must put my glasses on. I won't be able to see clearly. (unless)

e. He played until the end of the match. He had injured his leg. (although)

f. She was too late to meet her father at the airport. She waited for him at home. (consequently)

g. Road conditions were extremely bad. There were no accidents. (in spite of)

h. I can't afford a holiday this year. I've saved very little money. (as)

i. He stood on the chair. He could reach the top shelf of the bookcase. (so that)

j. They tied the thief up. He escaped. (despite)

3. Choose the correct word from the list.[xxii]

1) The device enables temperatures to be monitored, …….. improving the safety margin.

a. thereby
b. nevertheless
c. namely
d. e.g.

2) A new technique, …….. the infra-red camera, means that dust surrounding new stars can be penetrated.

a. whereas
b. namely
c. besides
d. for instance

3) Superconducting, …….. when materials lose all resistance, will boost computer performances.

a. thus
b. obviously
c. besides
d. that is to say

4) Programmable electronic systems are more reliable. …….. , they can be used to handle radio-active material.

a. Whereas
b. Such as
c. Moreover
d. In other words

5) Computer can process data extremely fast. …….. this, they have several serious drawbacks.

a. In spite of
b. Whereas
c. However
d. Moreover

6) The new engine is far more efficient. ……., more work is required to reduce noise levels.

a. Nevertheless
b. Whereas
c. i.e.
d. besides

7) The presence of high concentrations of elements that are rarely found on Earth, …….. iridium, suggests there was a meteorite impact at that spot.

a. on the whole
b. moreover
c. such as
d. that is to say

8) The data is stored on hard disk, …….. it is easily accessible.

a. actually
b. whereas
c. e.g.
d. hence

9) Applicants for the job should speak at least one other European language …….. French

a. obviously
b. besides
c. moreover
d. actually

10) ……., in the initial stages there were minor problems to be overcome, but from then on, the prototypes were perfectly reliable.

a. Besides
b. As a rule
c. Namely
d. despite

4. Write chains of linked sentences, using different linking words, each time based on one of the sentences below.[xxiii]

Example:

More women are getting executive jobs in high finance.

1. Clearly, this means that they are under greater stress.

➔ 2. In addition, women perform far more cognitive and emotional labour than men at home.

➔ 3. The number of women who need psychological help is therefore increasing.

1. Gold is a very expensive metal.

2. Death rates from infectious diseases will be reduced by 20% over the next 30 years.

3. The first atomic bomb was dropped in 1945.

4. Oil reserves will run out by the middle of the next century.

5. Women are now getting better results than men at university.

6. There is more and more traffic in town centres.

- in addition, furthermore, besides
- however, nevertheless, yet, in spite of, on the other hand
- but actually, but in reality, in fact
- obviously, clearly, doubtless
- in general, on the whole, as a rule
- in other words, that is to say, namely
  consequently, therefore, thus

Chain 1:

1.

 2.

 3.

Chain 2:

1.

 2.

 3.

VII - AUDIO: PROFESSOR HAWKING

A. LISTENING[xxiv]

- ***1st listening***

1. Listen to the recording. Write down as many link words from the audio as you can.

4. Take notes of anything you understand.

- ***2nd & 3rd listenings***

Answer the following questions.

1. What is Stephen Hawking's profession and what degrees did he have?

2. What did he suffer from?

3. What impact did it have on his life?

B. WRITING

Fill in the transcript with the missing connectors.

as a result of - as a rule - besides – despite - doubtless

- however - nevertheless - that is to say – thereby – i.e.

PROFESSOR STEPHEN HAWKING

Stephen Hawking is _____ the most well-known theoretical physicist and cosmologist of our time.

At the age of 21, while he was still a young student, Stephen Hawking was diagnosed as having amyotrophic lateral sclerosis, _____, a degenerative motor neurone disease that presents voluntary control of the muscles. He was informed that he did not have long to live.

_____ the disease, or as he himself says, perhaps _____ it, he continued with his studies and was awarded his PhD in 1977 and so started a brilliant scientific career. He has made important contributions to the theory of black holes and the origin of the universe.

_____ losing the ability to control his arm and leg muscles, he gradually lost the ability to speak clearly. This meant that, _____, at scientific conferences or seminars, he was obliged to speak via an interpreter.

In 1985, he caught pneumonia and was obliged to undergo tracheotomy, _____an incision of the oesophagus.

The tracheotomy operation, _____, removed his ability to speak altogether. For a time, the only way he could communicate was by signalling with his eye muscles.

A computer expert in California designed him a program linked to a voice synthesizer that could be activated with one finger, _____ enabling Hawking to move the cursor. At present, he can produce 15 words a minute.

Hawking is _____very active. He travels a great deal and recently has given conferences in the US, India, South Korea and the UK.

VIII - PRACTISE FOR YOUR PROJECT: CONNECTING THE DOTS

Instructions

1. Take at least 2 popular science articles (current research) from your project/ that could be used in your project.

2. Identify **5 linking words** (or expressions) in context between paragraphs or between ideas from 1-2 article(s), find synonyms and give a short explanation of their use.

e.g. Plants need sunlight to grow. <u>However</u>, some can survive in low light.

- *Linking word: however*
- *synonyms: nevertheless, yet*
- *explanation: used at beginning of a sentence / to express contrast*

/	Linking word (or expression)	Synonyms	Explanation of its use in context
1.			
2.			
3.			
4.			
5.			

Now, look at your two articles: What is the link between them? Check the section 'Vocabulary: Linking your ideas (general)' and think of an interesting research angle (addition, contrast, cause, consequence, concession, …) between articles. Fill in at least **two** tables. You should come back to this exercise later during the year and fill in more tables as you draw more **logical links between your articles**.

Link between article n°….	and article n°….

Link between article n°….	and article n°….

Link between article n°….	and article n°….

Link between article n°….	and article n°….

Link between article n°….	and article n°….

Link between article n°….	and article n°….

Link between article n°….	and article n°….

In class

1. Presenter: Present the gist of two of your current research articles and focus on the links between the articles. Use the analysis table provided in the following pages to help you better analyze your article (one table per article).

Do not read a text. Use a few notes only (keywords on a small piece of paper) and a PPT slide per article if you wish.

1. Listeners: Take notes of the information (choose one article per student) in the table below.
2. Be ready to report back on other students' current research articles. Double-check if the 2 articles are acceptable (popularized scientific articles, not news articles).

Current research article 1:	Current research article 2:
Current research article 3:	Current research article 4:
Current research article 5:	Current research article 6:

4. Discussion: Feedback about the presentations.

VIII – THE USE OF AI

The documents will be provided in-class.

PROJECT: 1st PROJECT CHECKPOINT

In groups of 3.

Each student should:

- explain their project (topic choice, …) for 2 minutes.
- explain how and why they selected their five articles about current research for 1 minute (no need for slides for this part)
- present **current research articles 1 & 2** with PowerPoint for 4 minutes. Use the analysis table provided in the following pages (one table per article) to help you. Don't forget to make links between your articles.



Use connectors.

The listeners act as the teacher. They should:

- ask why the student has chosen the topic.
- ask the student how and why they selected their current research articles
- ask questions about the presentation (e.g. What have you learnt while preparing this project?).
- give feedback about the presentation.
- give feedback about the PowerPoint.

See checklist next page.

CHECKLIST: STUDENT 1: …

Tick as appropriate.

1. PRESENTATION

Topic choice	☐ Clearly explains the topic choice ☐ 2 min
Article selection	☐ Explains how the five articles were selected ☐ Explains why these articles were chosen ☐ All selected articles are relevant ☐ 1 min
Presentation of research articles 1 & 2	☐ Presents Article 1 clearly (researchers, general topic, procedure/what was examined, results, and conclusion) ☐ Presents Article 2 clearly (researchers, general topic, procedure/what was examined, results, and conclusion) ☐ Uses connectors to link ideas smoothly ☐ Makes links between the 2 articles ☐ 4 min
Overall Delivery	☐ Speaks clearly ☐ Engages the audience (eye contact, posture,...)

2. POWERPOINT (to be examined after the presentation)

Slides	□ Front slide □ Table of contents □ General information □ Current research: 1 slide per article □ Bibliography
	□ Slide numbers
Front slide	□ Students' names (first name – SURNAME) □ Studies □ Academic year □ Title □ Teacher's name □ Course title
Overview	□ Title: Overview (NOT Summary/Table of Contents) □ No slide number next to each title
Content	□ Key words (NO text) □ Bullet points □ Relevant pictures only □ Current research part: for each article: title of article, name of researchers, date of publication, and useful info (e.g. general topic, procedure, conclusions, remaining questions…)

3. SPELLING/GRAMMAR

- Did you notice any typo on the slides?
- Any comments related to the language (grammar/voc)?

4. **ARE ALL THE SELECTED ARTICLES ACCEPTABLE? (popularized scientific articles NOT news articles)?**

Article 1:

Article 2:

Article 3:

Article 4:

Article 5:

CHECKLIST: STUDENT 2: …

Tick as appropriate.

1. PRESENTATION

Topic choice	☐ Clearly explains the topic choice ☐ 2 min
Article selection	☐ Explains <u>how</u> the five articles were selected ☐ Explains <u>why</u> these articles were chosen ☐ All selected articles are relevant ☐ 1 min
Presentation of research articles 1 & 2	☐ Presents Article 1 clearly (researchers, general topic, procedure/what was examined, results, and conclusion) ☐ Presents Article 2 clearly (researchers, general topic, procedure/what was examined, results, and conclusion) ☐ Uses connectors to link ideas smoothly ☐ Makes links between the 2 articles ☐ 4 min
Overall Delivery	☐ Speaks clearly ☐ Engages the audience (eye contact, posture,...)

2. POWERPOINT (to be examined after the presentation)

Slides	□ Front slide □ Table of contents □ General information □ Current research: 1 slide per article □ Bibliography □ Slide numbers
Front slide	□ Students' names (first name – SURNAME) □ Studies □ Academic year □ Title □ Teacher's name □ Course title
Overview	□ Title: Overview (NOT Summary/Table of Contents) □ No slide number next to each title
Content	□ Key words (NO text) □ Bullet points □ Relevant pictures only □ Current research part: for each article: title of article, name of researchers, date of publication, and useful info (e.g. general topic, procedure, conclusions, remaining questions…)

3. SPELLING/GRAMMAR

- Did you notice any typo on the slides?
- Any comments related to the language (grammar/voc)?

4. **ARE ALL THE SELECTED ARTICLES ACCEPTABLE? (popularized scientific articles NOT news articles)?**

Article 1:
Article 2:
Article 3:
Article 4:
Article 5:

CHECKLIST: STUDENT 3: …

Tick as appropriate.

1. PRESENTATION

Topic choice	☐ Clearly explains the topic choice ☐ 2 min
Article selection	☐ Explains how the five articles were selected ☐ Explains why these articles were chosen ☐ All selected articles are relevant ☐ 1 min
Presentation of research articles 1 & 2	☐ Presents Article 1 clearly (researchers, general topic, procedure/what was examined, results, and conclusion) ☐ Presents Article 2 clearly (researchers, general topic, procedure/what was examined, results, and conclusion) ☐ Uses connectors to link ideas smoothly ☐ Makes links between the 2 articles ☐ 4 min
Overall Delivery	☐ Speaks clearly ☐ Engages the audience (eye contact, posture,...)

2. POWERPOINT (to be examined after the presentation)

Slides	□ Front slide □ Table of contents □ General information □ Current research: 1 slide per article □ Bibliography □ Slide numbers
Front slide	□ Students' names (first name – SURNAME) □ Studies □ Academic year □ Title □ Teacher's name □ Course title
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Content	□ Key words (NO text) □ Bullet points □ Relevant pictures only □ Current research part: for each article: title of article, name of researchers, date of publication, and useful info (e.g. general topic, procedure, conclusions, remaining questions…)

3. SPELLING/GRAMMAR

- Did you notice any typo on the slides?
- Any comments related to the language (grammar/voc)?

4. **ARE ALL THE SELECTED ARTICLES ACCEPTABLE? (popularized scientific articles NOT news articles)?**

Article 1:
Article 2:
Article 3:
Article 4:
Article 5:

PROJECT: CURRENT RESEARCH: EXPRESSIONS & ANALYSIS TABLE

RESEARCHERS & TOPIC

- This research study was carried out by X from Y University in (*country)*. It investigates (*the topic)*.
  conducted

Scientists Researchers X	from Y Universit y	(have)	carried out conducted	a study	on *(the topic*). about

PUBLICATION

- It was published in *(name of the journal)* in (January) 2025.
- They published their work in "
- The work appeared in "

PROCEDURE / WHAT WAS EXAMINED

- ***Sequence***

First, Second, Third, …	First of all, Then, Finally,	The first step was The second step " The third step " …

- ***Verbs***

The	researchers scientists team	tested … carried out an experiment about/on … analysed… observed… examined… claimed that… reported that… said that…

CONCLUSIONS

- The	researchers scientists team	(have)	found (evidence) demonstrated discovered	that …

- This suggests that…

- The	results findings	showed suggested	that…

- The	researchers scientists team	(now) believe concluded found out discovered	that…

- The	researchers' scientists' team's	discovery	implies that…

REMAINING QUESTIONS

- Several challenges remain including how…

- Further studies will focus on + noun / -ing verb

- It is still unclear how…

ANALYSIS TABLE: ARTICLE 1:……..

Researchers?	
General topic **(Introduction: State of the art / Background information)**	
Procedure/what was examined and how? **(Material & Methods)**	
Results / what was found?	
Conclusion/Discussion/ Prospects	

ANALYSIS TABLE: ARTICLE 2: ……..

Researchers?	
General topic **(Introduction: State of the art / Background information)**	
Procedure/what was examined and how? **(Material & Methods)**	
Results / what was found?	
Conclusion/Discussion/ Prospects	

MOCK EXAM: GRAMMAR

I - Choose either the *present simple* or the *present continuous* verb tense and enter the correct form.

A Trekking Journal

November 12, 2013

Today (be) ……1 the second day of my trek around Mount Annapurna. I am exhausted and my legs (shake) ……2; I just hope I am able to complete the trek. My feet (kill, really) ……..3 me and my toes (bleed) ……4, but I (want, still) ……5 to continue.

Nepal is a fascinating country, but I have a great deal to learn. Everything (be) ……6 so different, and I (try) ……7 to adapt to the new way of life here. I (learn) ……..8 a little bit of the language to make communication easier; unfortunately, I (learn, not) ……..9 foreign languages quickly. Although I (understand, not) ……10 much yet, I believe that I (improve, gradually ……..11.

I (travel, currently) ……12 with Liam, a student from Leeds University in England. He (be) ……13 a nice guy, but impatient. He (walk, always) ……14 ahead of me and (complain) ……15 that I am too slow. I (do) ……..16 my best to keep up with him, but he is younger and stronger than I am. Maybe, I am just feeling sorry for myself because I am getting old.

Right now, Liam (sit) ……17 with the owner of the inn. They (discuss) ……..18 the differences between life in England and life in Nepal. I (know, not) ……19 the real name of the owner, but everybody (call, just) ……20 him Tam. Tam (speak) ……21 English very well and he (try) ……..22 to teach Liam some words in Nepali. Every time Tam (say) ……..23 a new word, Liam (try) ……24 to repeat it. Unfortunately, Liam (seem, also) ……25 to have difficulty learning foreign languages. I just hope we don't get lost and have to ask for directions.

II - Choose either the *past simple* or the *present perfect* verb tense and enter the correct form.

A: Did you try the new coding framework?
B: I don't know. I (use, never) ……1 it before.

Sam (submit) ……2 his project to GitHub last week.
My best friend and I (know) ……3 each other since our first year in Computer Science. We still collaborate on projects once a month.

Dr. Lin is an amazing programmer. She (develop) ……4 ten open-source libraries this year. One day, she'll be as famous as Guido van Rossum.

I (experience, not) ……5 a bug this frustrating since I (start) ……6 learning Python.

Things (change) ……7 at TechLabs, Inc. When we first (join) ……8 the team three years ago, the company (have, only) ……9 fifteen developers. Since then, we (expand) ……10 to include over 500 employees.

I (advise) ……11 him to follow the coding guidelines, but he (ignore) ……12 them and (introduce) ……13 several bugs into the project.

Listen Donna, I don't care if you (miss) ……14 the team meeting this morning. You (be) ……15 late to stand-ups too many times. You are removed from the project!

Sam is from Colorado, which is far from Silicon Valley, so he (attend, never) ……16 a major tech conference. He should come with us to San Francisco.

How frustrating! George (dream) ……17 of presenting at a big AI conference, but he didn't manage it. He (attend, never) ……18 one yet.

In the last twenty years, programming (become) ……19 much easier and more accessible. In the 1990s, it (take) ……20 months to set up servers and deploy applications. The process (be) ……21 very slow and often frustrating. Things (change) ……22 a great deal in the last two decades. Now you can deploy apps to the cloud in a matter of hours.

Johnny, I can't believe how much you (change) ……23 since the last time I (see)……24 you.

III - Use the verbs in brackets in the correct future tenses – will-future, going to-future, Simple Present or Present Progressive.

1. The new software update _____ at 10:00 tomorrow morning. (to be released)
2. We _____ a team meeting to discuss the project soon. (to have)
3. It _____ a major security patch next week. (to require)
4. Look at the servers – they _____ in a few minutes. (to reboot)
5. On Sunday at 8 o'clock I _____ my mentor online. (to meet)
6. They _____ to San Francisco for the tech conference on Friday. (to fly)
7. Wait! I _____ you to the office for the presentation. (to drive)
8. I _____ my lab partner in April to test the new algorithm. (to see)
9. Listen! There's a message on Slack. I _____ it for you. (to check)

IV - The following sentences are answers to questions. If the words in bold are the missing information, write the questions.

1. He codes in Python **twice a week**.

_____

2. I have debugged the problem **five times**.

_____

3. Hua Hin is **200 kilometres** from Bangkok.

_____

4. She bought **a new laptop**.

_____

5. There were **more than 1,000 molecules** in the solution.

_____

6. Kate had to wait **an hour** for the meeting to start.

_____

7. Tom likes **JavaScript**.

_____

8. The new workstation cost **$10,000**.

_____

9. The project starts **this week**.

_____

10. Mary joined the tech company **in 2011**.

_____

11. John met his girlfriend **at a hackathon**.

_____

12. They are staying **at the university dormitory**.

_____

13. He paid for the softwares licenses with **his credit card**.

_____

14. Kim is **160 centimetres** tall.

_____

MOCK EXAM: VOCABULARY

Find the missing word(s) in context. The first letter as well as a synonym, antonym or an explanation of the word is given to help you.

1. Between 2005 and 2025, the average world death rate is expected to decline only s…….. (a little, by a small amount)

2. The biosphere refers to the t……. layer surrounding the Earth where living organisms are found. (small transversal dimension, >< thick)

3. The original estimation of the cost of building the Three Gorges Dam in China was i…… . (imprecise)

4. A chain reaction is a series of reactions in which the result of each s……. causes the next. (step, part of the process)

5. If you add two odd numbers together, you get an e……. number. (>< odd)

6. The Arctic Ocean is joined to the Pacific Ocean by the Bering Strait, a narrow and s……. channel with a depth of no more than 55 m. (>< deep)

GRAMMAR: HOW TO ASK QUESTIONS

I - DISCOVER

1. CLOSED QUESTIONS: YES-NO QUESTIONS

It is here → _____

He studies Pharmacy. → _____

He has got a new laptop. → _____

They went to UMONS. → _____

RULE : yes-no questions

- ***Single verb (present simple or past simple)***
  - 'to be' : to be conjugated + subject + complements
  - other verbs: auxiliary "do" + subject + infinitive + complements
    does/do/did

- ***Compound verb (more than one part:)***

  with perfect forms : auxiliary "have" conjugated + subject + past participle + compl
  3 rd colomn

  with progressive forms : to be conjugated + subject + gerund (regular verbs)(v-ing) + compl

  with modals : modal + subject = infinitive

2. OPEN QUESTIONS: WHO - WHAT - WHY - WHEN - WHERE - HOW– HOW MUCH - HOW MANY - WHOSE

Fleming is a Scottish scientist. → who is fleming

Fleming discovered penicillin in Edinburgh. → is she student

William loves Kate. → does it snow ?

William loves Kate. → do I speak spanish ?

She has requested an appointment → do I go to gent yesterday

It is Mark's car. → am I go what has she requested

→ should whose car is it

RULE : open questions

- ***Single verb (present simple or past simple)***
  - 'to be' : WH-word + to be conjugated + subject + compl
  - other verbs: WH-word + auxiliary "do" + subject+ infinitive + compl



Exception: If the subject is WHAT or WHO - *e.g. William loves Kate.*

WHO/WHAT + verb conjugated+ compl

- ***Compound verb (more than one part)***

WH-word +

with perfect forms : wh + auxiliary "have" conjugated + subject + past participle + compl

with progressive forms : wh + to be conjugated + subject + gerund (regular verbs)(v-ing) + compl

with modals : wh + modal + subject = infinitive

Note: Whose + ... prepositions are always placed at the end of the question

II - THEORY

WORD ORDER[3]

In questions, the word order *subject-verbs-object* is the same as in affirmative sentences. The only thing that's different is that you usually have to put the auxiliary verb (or the main verb "be") before the subject. Interrogatives are put at the beginning of the sentences:

interrogative	***auxiliary verb***	***subject***	***other verb(s)***	***indirect object***	***direct object***	***place***	***time***
What	would	you	like to tell	me			
	Did	you	have		a party	in your flat	yesterday?
When	were	you				here?	

You don't use an auxiliary verb if you ask for the subject. In this case the interrogative simply takes the place of the subject.

interrogative	***verb(s)***	***object***
Who	asked	you?

Other useful website: http://www.englisch-hilfen.de/en/grammar/frage.htm

[3] *Source : http://www.ego4u.com/en/cram-up/grammar/word-order/questions*

QUESTION WORDS[4]

We use question words to ask certain types of questions. We often refer to them as *WH words* because they include the letters *WH* (for example *WHy, HoW*).

question word	***function***	***example sentence***
what	asking for information about something	What is your name?
	asking for repetition or confirmation	What? I can't hear you. You did what?
what...for	asking for a reason, asking why	What did you do that for?
when	asking about time	When did he leave?
where	asking in or at what place or position	Where do they live?
which	asking about choice	Which colour do you want?
who	asking what or which person or people (subject)	Who opened the door?
whom	asking what or which person or people (object)	Whom did you see?
whose	asking about ownership	Whose are these keys? Whose turn is it?
why	asking for reason, asking what...for	Why do you say that?

[4] *Source:* https://www.englishclub.com/vocabulary/wh-question-words.htm

why don't	making a suggestion	Why don't I help you?
how	asking about manner	How does this work?
	asking about condition or quality	How was your exam?
how + adj/adv	asking about extent or degree	see examples below
how far	distance	How far is Pattaya from Bangkok?
how long	length (time or space)	How long will it take?
how many	quantity (countable)	How many cars are there?
how much	quantity (uncountable)	How much money do you have?
how old	age	How old are you?
how come (informal)	asking for reason, asking why	How come I can't see her?

QUESTION TAGS

They are mini-questions used at the end of the sentence to :

1. check whether the sentence is true or not ;
2. encourage a reply.

A. Positive sentence

=> negative tag = auxiliary in the negative form + pronoun.

- ***You are*** *happy*.



You aren't *happy.*



You aren't



Aren't you

You are happy, ***aren't you ?***

- *You have a cat,* ***don't you****?*
  *(****You don't*** *have a cat.)*
- *It has rained a lot,* ***hasn't it*** *?*
  *(****It hasn't*** *rained a lot.)*
- *This car costs a lot,* ***doesn't it*** *?*
  (This car = ***It doesn't*** *cost a lot.)*

B. <u>Negative sentence</u>

=> positive tag = auxiliary in the positive form + pronoun.

- ***You aren't*** *happy.*



You are *happy.*



You are



Are you

You aren't happy, ***are you ?***

- *You don't have a cat,* ***do you****?*
  *(****You do*** *have a cat.)*
- *It hasn't rained a lot,* ***has it*** *?*
  *(****It has*** *rained a lot.)*

- *This car doesn't cost a lot, **does it** ?*
  *(This car = **It does** cost a lot.)*

C. Exceptions

- *Don't mention it, **will you** ?* (Imperatives)
- *Let's talk about it, **shall we** ?* (Let's = Let us)
- *Sam is never on time, **is he** ?* (Never is negative !)

SHORT ANSWERS

Short answers are brief but complete answers to yes/no questions.

A. Positive

(Yes,) subject + auxiliary in the correct tense (positive)

- ***Is she** happy ?* *Yes, **she is**.*
- ***Do you** have a cat ?* *Yes, **I do**.*
- ***Has it** rained ?* *Yes, **it has**.*
- ***Does** this car (=**it**) cost a lot ?* *Yes, **it does**.*

B. Negative

(No,) subject + auxiliary in the correct tense (negative)

- ***Is she** happy ?* *No, **she isn't**.*
- ***Do you** have a cat ?* *No, **I don't**.*
- ***Has it** rained ?* *No, **it hasn't**.*
- ***Does** this car (=**it**) cost a lot ?* *No, **it doesn't**.*

C. Exceptions

	Positive	Negative
• *Is she happy ?*	*I think **so***	*I **don't** think **so***
• *Is she happy ?*	*I hope **so***	*I hope **not***
• *Is she happy ?*	*I suppose **so***	*I suppose **not***

Other useful website: http://www.anglaiscours.fr/poser-une-question-en-anglais-what-where-who-why-how-whose-when-which-etc.html

III – EXERCISES

1. Yes-no questions: Transform the following sentences into questions.

a. She is a student → is she student

b. It's snowing → is it snowing ?

c. I speak Spanish → ~~do I speak spanish ?~~ do you go to gent yesterday

d. I went to Gent yesterday → am I go

e. I'm going to the party → should I study more ?

f. I should study more → will I try to come

g. I will try to come → ~~has she gone to NY~~

h. She has gone to NY → _____

2. Open questions: Ask questions with *what, why, when, where, how, whose...* The answers are underlined.

a. She is at university <u>during the week</u> → _____

b. The students are i<u>n building 6</u> → _____

c. I am leaving <u>at 6</u> → ~~when are you leaving~~

d. I am at university <u>to study science</u> → _____

e. She is feeling <u>terrible</u> → _____

f. I am studying <u>my physics lesson</u> → ~~whare~~

g. He starts to work <u>at 8 am</u> → _____

h. He studies <u>in Mons</u> → _____

i. He lived <u>in Brussels</u> for 3 years → ~~where did he live~~

j. He gave a presentation <u>yesterday</u> → when did he give

k. She has gone <u>to London</u> → _____

l. <u>He should study more</u> → _____

m. Paul knows <u>Virginia</u> → _____

n. <u>Paul</u> knows Virginia → _____

o. It is <u>Steve's</u> house → _____

3. Ask questions. The answers are the words in bold. Please keep all relevant information in the question.

1. The students have prepared **a presentation about an organ**.

_____?

2. He went **to the hill**.

Saisissez du texte ici

_____?

3. They regularly cut **the grass**.

_____?

4. **He** has given Harry a nice present.

_____?

5. The guests made **an awful mess**.

_____?

6. This cave is around **26 meters** in depth.

_____?

(~~What is the depth….~~)

7. They went home **because they were tired**.

_____?

8. We call Mary **every weekend**.

_____?

9. He stays **in bed** all day long.

_____?

10. My suitcase weighs **24 kilogrammes**.

_____?

11. He has stayed in London **for a few hours**.

_____?

12. She wore black jeans **last Tuesday**.

_____?

13. Tina has lived in a small village in Wales **since she was a little girl**.

_____?

14. They went up **1,860** steps to reach the top.

_____?

15. We took **the second elevator**, because the first one was broken.

_____?

5. Fill in the blanks with a question-tag

1. It freezes every night now, _____ ?
2. She'd eaten, _____ ?
3. Don't lose it, will you ?
4. He's become a doctor, hasn't he ?
5. They'll never agree, will they ?
6. They are never on time, are they ?
7. You'd help her if you could, wouldn't you ?
8. They haven't helped you, have they ?
9. You didn't lie, did you ?
10. Let's get going, shall ?
11. Jim broke the expensive machine, didn't him ?
12. She will come, won't she ?
13. We'd never been to Rome before, had we ?
14. There is a stadium in that town, isn't there ?
15. It is snowing again, isn't it ?
16. Let's have a talk, sha ?
17. Wait a minute, will you ?
18. He had to go there, didn't ?
19. You aren't coming, are you ?
20. He wasn't your friend, was he ?

6. Answer the questions using a short answer

1. Can you swim ? No, cat .

2. Will he be there ? No, _____ .
3. Have you got a car ? Yes, _____ .
4. Have you been running ? Yes, _____ .
5. Who will be there ? (Peter) _____ .
6. Who should come ? (me) _____ .
7. Who doesn't like tea ? (my friend) _____ .

GRAMMAR: LIST OF TENSES

Tense	Active	Passive
Present simple	*I watch.*	*I am watched.*
Present continuous	*I am watching.*	*I am being watched.*
Present perfect simple	*I have watched.*	*I have been watched.*
Present perfect continuous	*I have been watching.*	-
Past simple	*I watched.*	*I was watched.*
Past continuous	*I was watching.*	*I was being watched.*
Past perfect simple	*I had watched.*	*I had been watched.*
Past perfect continuous	*I had been watching.*	-
Be going to	*I am going to watch him.*	*He is going to be watched (by me).*
Future simple	*I will watch.*	*I will be watched.*
Future continuous	*I will be watching.*	-
Future perfect simple	*I will have watched.*	*I will have been watched.*
Future perfect continuous	*I will have been watching.*	-
Imperative	*Watch !*	*Be reassured !*
Conditional present	*I would watch.*	*I would be watched.*

Match the tense and its form.

Present simple	•	• I am studying
Present continuous	•	• I have been studying
Present perfect simple	•	• I studied
Present perfect continuous	•	• I had studied
Past simple	•	• I had been studying
Past continuous	•	• I study
Past perfect simple	•	• I have studied
Past perfect continuous	•	• I was studying

GRAMMAR: FUTURE

	WILL	BE GOING TO	PRESENT CONTINUOUS
PREDICTION	▪ **Calculating** *e.g. I think it will rain this evening.* (calculating, weather forecast) ▪ **We think, we believe** *e.g. Don't take that glass. You'll break it.* (I know you, I think you are very clumsy – Reformulation: If you take that glass, you will break it)	▪ **Future starting now** *e.g. I think it is going to rain this evening.* (I can see clouds in the sky) *e.g. Look out! You are going to break that glass!* (There is an obstacle, you are going to fall or to drop it… I can see it starting to happen now)	
PLAN, INTENTION	**Social arrangement** *e.g. I'll see you tomorrow*	**Plans (arrangements may or may not have been fixed)** *e.g.* *. I am going to speak at conference. (no fixed arrangement)* *. I am going to speak at conference on 15 Dec. (fixed arrangement)* *. We are going to go to Paris next year.*	**Plans with already <u>fixed</u> arrangements – the time / place is given** *e.g.* *. I am speaking at conference on 15 Dec.* *. We are going to Paris next year.*
DECISION	**Making the (quick) decision now** e.g. *. We have got a lot of mail to answer. - OK, I'll take care of it* (I am making the quick decision now) *. Which one do you want? I'll take this one.*	**The decision was made earlier** *e.g. We have got a lot of mail to answer. - I know. I'm going to answer them on Tuesday.* (I decided a while ago that I would answer the mail on Tuesday)	
PROMISE	*e.g. Please write every week – Of course, I will (I promise)*		
THREAT	*e.g. Do your homework or I'll take your playstation from you.*		
REFUSAL	*e.g. The car won't start.*		
OFFER	*e.g. I'll carry your bag.*		

PRESENT SIMPLE	Event always happens a certain time *e.g. Her plane arrives at 15.30.* Or Event is in the time table *e.g. The lesson starts at 10 am.*

Choose the best form.

1. It *will rain/is going to rain* – Look at those clouds.
2. If it gets any colder, it *will snow/is going to snow*.
3. Alice *will have / is going to have* a baby.
4. If the baby takes after his mum, he'*ll have/is going to have* blue eyes.
5. I *am going to play / am playing* with Stan this Saturday.
6. *He'll win/he's winning* tomorrow. He always does.
7. Don't tell her. She'*ll tell/is going to tell* everybody else.
8. She'*ll get married/she's getting married* on Friday at the local church.

GRAMMAR: PAST AND PERFECT TENSES

	PAST SIMPLE	PAST PROGRESSIVE
Use	Action completely in the past, dated by a time marker or through context	Action happening in the past and interrupted by another action
Affirmative form	➢ Regular verbs: infinitive + -ED *e.g. Yesterday, I played tennis.* ➢ Irregular verbs: to be studied by heart *e.g. In 2015, I went to Scotland.*	WAS/WERE + ING-verb *e.g. I was watching TV when the phone rang.*
Interrogative form	DID + subject + infinitive *e.g. Did you visit Scotland last year?*	WAS/WERE + subject + ING-verb *e.g. Were you watching TV when the phone rang?*
Negative form	DIDN'T + infinitive *e.g. I didn't visit Scotland last year.*	WASN'T / WEREN'T + ING-verb *e.g. I wasn't watching TV when the phone rang.*

	PRESENT PERFECT SIMPLE	PAST PERFECT SIMPLE
Use	➢ Action that started in the past and goes up to now ➢ Finished actions that have consequences now ➢ News items	Action that happened before another past action
Affirmative form	HAVE/HAS + past participle *e.g.* *. I have lived in Mons since 2015.* *. I have eaten all the cakes (there are none left).* *. A plane has crashed in Bangkok.*	HAD + past participle *e.g.* *. I bought a train ticket and went to the platform. But the train had already left.* *. It was a song I hadn't heard before.* *. I went into the office. It was empty. Natalie had gone home.*
Interrogative form	HAVE/HAS + subject + past participle *e.g. Have you done it?*	HAD + subject + past participle *e.g. Had he finished his exam when the bell rang?*
Negative form	HAVEN'T/HASN'T + past participle *e.g. I haven't been to Paris for a long time.*	HADN'T+ past participle *e.g. I hadn't finished my exam when the bell rang.*

Identify the tense of the verbs and justify its use.

1. I have studied for 6 hours.
2. I have been studying for 6 hours.
3. Yesterday, I studied for 6 hours.

4. I've had two cups of tea this morning.
5. I had two cups of tea this morning.

6. I was having a bath when the bell rang.
7. The bell rang while I was having a bath.

8. When I arrived, everybody applauded.
9. When I arrived, everybody was applauding.

10. After he finished watching the film, he went to bed.
11. After he had finished watching the film, he went to bed.

12. I was walking the dog when I saw my neighbour.
13. I had already walked the dog when I saw my neighbour.

GRAMMAR: LIST OF IRREGULAR VERBS

Infinitive	Past tense	Past participle	Translation
To beat	beat	beaten	battre
To become	became	become	devenir
To begin	began	begun	commencer
To bite	bit	bitten	mordre
To blow	blew	blown	souffler
To break	broke	broken	briser
To bring	brought	brought	apporter
To build	built	built	bâtir
To burn	burnt / burned	burnt / burned	brûler
To buy	bought	bought	acheter
To catch	caught	caught	attraper
To choose	chose	chosen	choisir
To come	came	come	venir
To cost	cost	cost	coûter
To cut	cut	cut	couper
To deal	dealt	dealt	distribuer, trafiquer
To do	did	done	faire
To draw	drew	drawn	tirer, dessiner
To dream	dreamt / dreamed	dreamt / dreamed	rêver
To drink	drank	drunk	boire
To drive	drove	driven	conduire, pousser
To eat	ate	eaten	manger
To fall	fell	fallen	tomber
To feed	fed	fed	se nourrir
To feel	felt	felt	(se) sentir, éprouver
To fight	fought	fought	se battre, combattre
To find	found	found	trouver
To fly	flew	flown	voler, s'envoler
To forbid	forbad(e)	forbidden	défendre, interdire
To forget	forgot	forgotten	oublier
To forgive	forgave	forgiven	pardonner
To freeze	froze	frozen	geler
To get	got	got	obtenir
To give	gave	given	donner
To go	went	gone	aller
To grow	grew	grown	grandir, pousser
To hang	hung	hung	pendre, suspendre
To hear	heard	heard	entendre
To hide	hid	hidden	(se) cacher
To hit	hit	hit	frapper, toucher
To hold	held	held	tenir
To hurt	hurt	hurt	blesser, faire mal

To keep	kept	kept	garder
To know	knew	known	savoir, connaître
To lay	laid	laid	poser, coucher
To lead	led	led	mener, conduire
To learn	learnt / learned	learnt / learned	apprendre
To leave	left	left	quitter, partir
To lend	lent	lent	prêter
To let	let	let	laisser, permettre
To lie	lay	lain	être couché
To lose	lost	lost	perdre
To make	made	made	faire, fabriquer
To mean	meant	meant	signifier
To meet	met	met	(se) rencontrer
To pay	paid	paid	payer
To put	put	put	poser, mettre
To read	read	read	lire
To ride	rode	ridden	aller à (cheval, vélo,…)
To ring	rang	rung	sonner
To rise	rose	risen	se lever
To run	ran	run	courir
To say	said	said	dire
To see	saw	seen	voir
To sell	sold	sold	vendre
To send	sent	sent	envoyer
To set	set	set	poser, fixer
To shake	shook	shaken	secouer
To shine	shone	shone	briller
To show	showed	shown / showed	montrer
To shut	shut	shut	fermer
To sing	sang	sung	chanter
To sink	sank	sunk	couler, sombrer
To sit	sat	sat	être assis
To sleep	slept	slept	dormir
To smell	smelt	smelt	sentir (odorat)
To speak	spoke	spoken	parler
To spend	spent	spent	passer, dépenser
To spread	spread	spread	(s')étendre
To stand	stood	stood	se tenir debout
To steal	stole	stolen	voler, dérober
To stick	stuck	stuck	coller
To strike	struck	struck	frapper
To swear	swore	sworn	jurer
To swim	swam	swum	nager
To take	took	taken	prendre
To teach	taught	taught	enseigner
To tear	tore	torn	déchirer

is she student
does it snow ?
do I speak spanish ?
do I go to gent yesterday
am I go
should

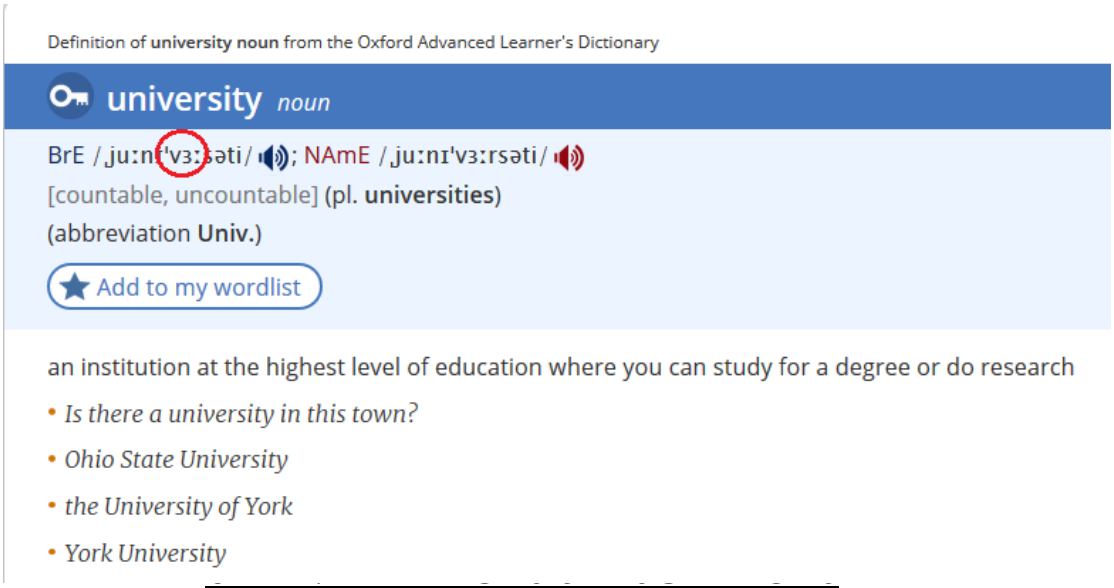
To tell	told	told	dire, raconter
To think	thought	thought	penser
To throw	threw	thrown	jeter
To understand	understood	understood	comprendre
To wake	woke	woken	éveiller
To wear	wore	worn	porter
To win	won	won	gagner
To write	wrote	written	écrire

APPENDIX 1: PRONUNCIATION: WORD STRESS

I - HOW CAN YOU FIND OUT WHERE THE STRESS GOES?

Look up the dictionary. There is an apostrophe before the stressed syllable in phonetic script. This syllable is pronounced a little longer and louder than the others.

is she student
does it snow ?
do I speak spanish ?
do I go to gent yesterday
am I go
shoul



II – OBSERVE AND DISCOVER SOME RULES

1. Listen and underline the stressed syllable in each word in the table below.

2 syllables	*3 syllables*
science physics chemist purpose process to process to confirm graphic data	scientist computer physicist measurement frequency consequence logical chemical integer
4 syllables	***5 syllables***
scientific mathematics biologist informatics comparison hypothesis modality centimetre to popularize	mathematician mathematical modification hypothetical pronunciation biological

2. Can you guess some of the pronunciation rules? Underline or tick the correct answer.

a. Most **2-syllable nouns** have the stress on the 1st – 2nd syllable.

b. Words ending in **–ic** have the stress on the □ antepenultimate syllable (3rd from the end)

□ penultimate syllable (2nd from the end)

□ last syllable

c. Words ending in **–tion** have the stress on the □ antepenultimate syllable (3rd from the end)

□ penultimate syllable (2nd from the end)

□ last syllable

d. Words ending in **–al** have the stress on the □ antepenultimate syllable (3rd from the end)

□ penultimate syllable (2nd from the end)

APPENDIX 2: PRONUNCIATION: WORD STRESS RULES[xxv]

Go to https://www.englishclub.com/pronunciation/word-stress.htm to find this information with sound files.

There are two very simple rules about word stress:

1. **One word has only one stress.** (One word cannot have two stresses. If you hear two stresses, you hear two words. Two stresses cannot be one word. It is true that there can be a "secondary" stress in some words. But a secondary stress is much smaller than the main [primary] stress, and is only used in long words.)
2. **We can only stress vowels, not consonants.**

Here are some more, rather complicated, rules that can help you understand where to put the stress. But do not rely on them too much, because there are many exceptions. It is better to try to "feel" the music of the language and to add the stress naturally.

I - STRESS ON FIRST SYLLABLE

Rule	***Example***
Most **2-syllable nouns**	PRESent Export CHIna TAble
Most **2-syllable adjectives**	PRESent SLENder CLEVer HAPpy

II - STRESS ON LAST SYLLABLE

Rule	***Example***
Most **2-syllable verbs**	preSENT exPORT deCIDE beGIN

There are many two-syllable words in English whose meaning and class change with a change in stress. The word **present**, for example is a two-syllable word. If we stress the first syllable, it is a noun (gift) or an adjective (opposite of absent). But if we stress the second syllable, it becomes a verb (to offer). More examples: the words **export**, **import**, **contract** and **object** can all be nouns or verbs depending on whether the stress is on the first or second syllable.

III - STRESS ON PENULTIMATE SYLLABLE (penultimate = second from end)

Rule	***Example***
Words ending in **-ic**	GRAPHic geoGRAPHic geoLOGic
Words ending in **-sion** and **-tion**	teleVIsion reveLAtion

For a few words, native English speakers don't always "agree" on where to put the stress. For example, some people say **teleVIsion** and others say **TELevision**. Another example is: **CONtroversy** and **conTROversy**.

IV - STRESS ON ANTE-PENULTIMATE SYLLABLE (ante-penultimate = third from end)

Rule	***Example***
Words ending in **-cy**, **-ty**, **-phy** and **-gy**	deMOcracy dependaBIlity phoTOgraphy geOLogy
Words ending in **-al**	CRItical geoLOgical

V - COMPOUND WORDS (words with two parts)

Rule	***Example***
For compound **nouns**, the stress is on the **first** part	BLACKbird GREENhouse
For compound **adjectives**, the stress is on the **second** part	bad-TEMpered old-FASHioned
For compound **verbs**, the stress is on the **second** part	underSTAND overFLOW

APPENDIX 3: PRONUNCIATION AND GRAMMAR: SIMPLE PAST[xxvi]

1. Listen to the following technical and scientific verbs in the simple past and put them in the right column.

analysed	*experimented*	*poured*	*solved*
boiled	*filled*	*processed*	*sprayed*
calculated	*guided*	*reached*	*touched*
caused	*heated*	*reacted*	*used*
contained	*improved*	*risked*	*warned*
crushed	*infected*	*searched*	*worsened*
cured	*limited*	*solidified*	

-ed pronounced /d/	***-ed pronounced /t/***	***-ed pronounced /id/***

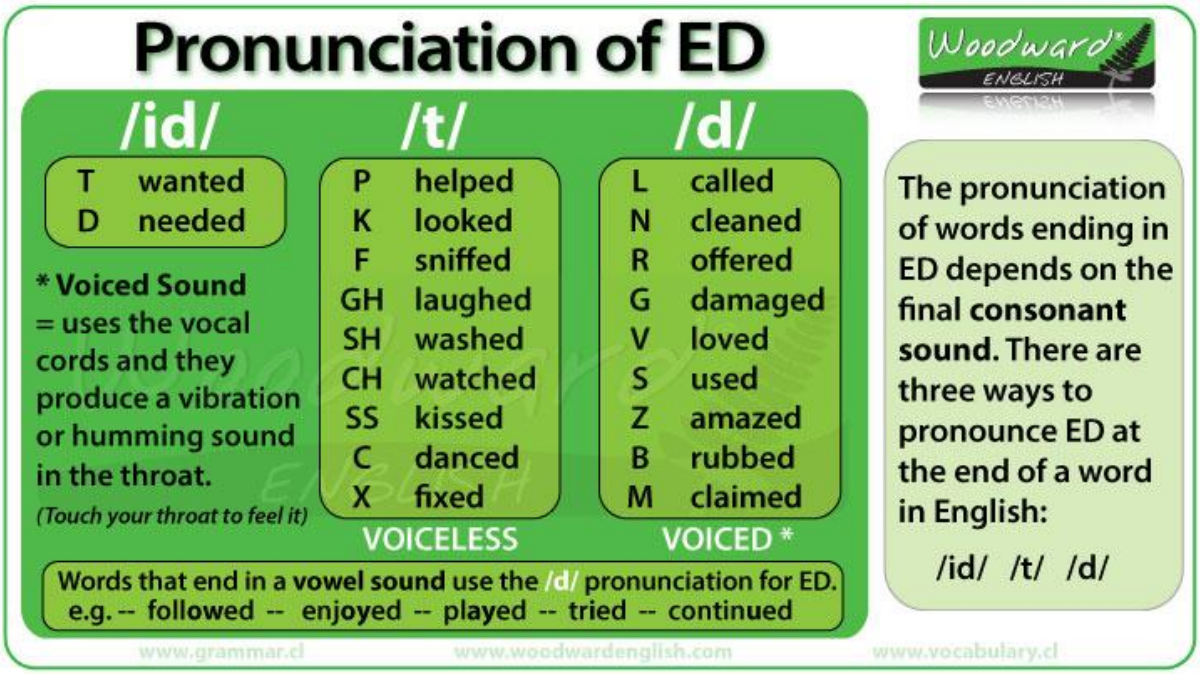
2. Can you guess the pronunciation rules?

a. After unvoiced sounds such as **p**, **f**, **s**, **k**, ʃ, and tʃ, the final ***–ed*** is pronounced like ____

b. For verbs ending in ***d*** and ***t,*** the final ***–ed*** is pronounced _____

c. After voiced sounds such as b, g, v, z, ʒ, dʒ, m, n, ŋ, l, ð, and r, the final *–ed* sound is pronounced like _____

Pronunciation of ED

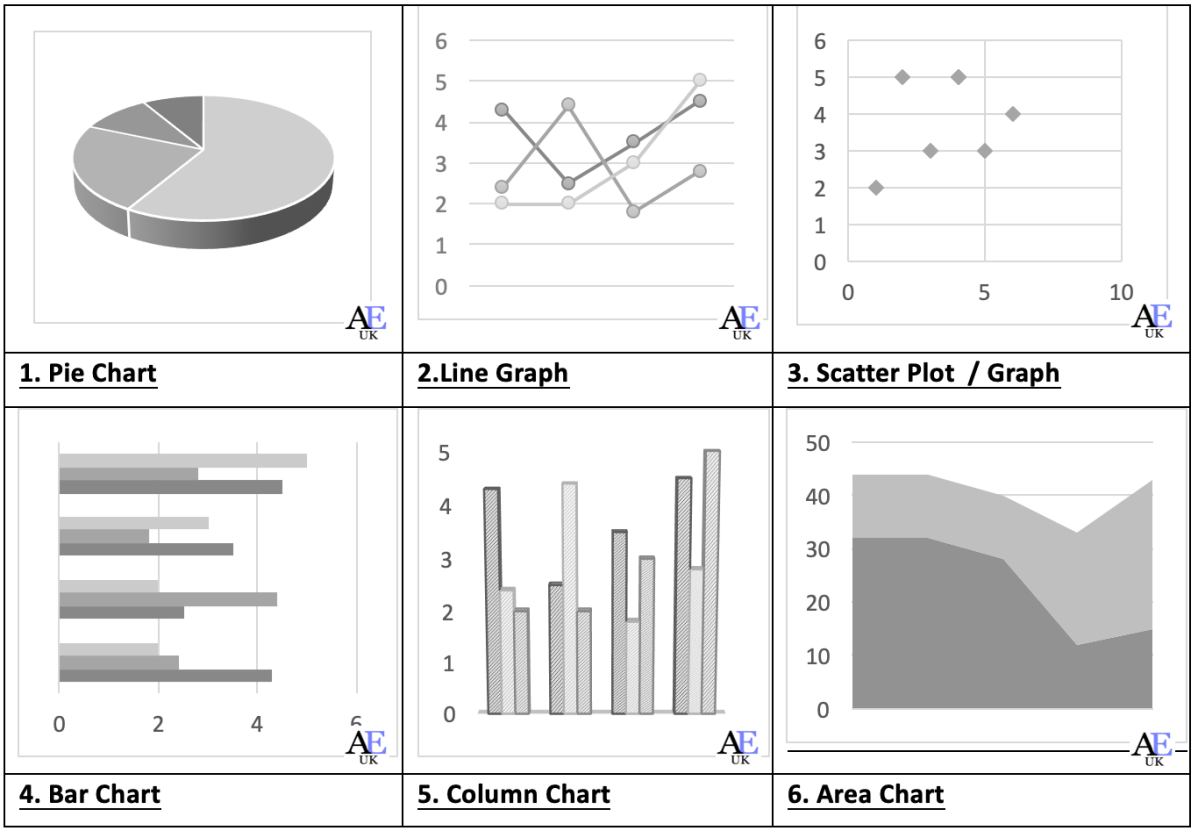


The pronunciation of words ending in ED depends on the final **consonant sound.** There are three ways to pronounce ED at the end of a word in English:

/id/ /t/ /d/

APPENDIX 4: DESCRIBING A GRAPH[xxvii]

TYPES OF GRAPHS



1. Pie Chart	2.Line Graph	3. Scatter Plot / Graph
4. Bar Chart	**5. Column Chart**	**6. Area Chart**

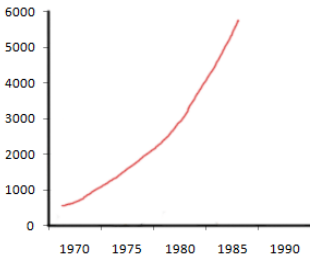
INTRODUCING THE GRAPH

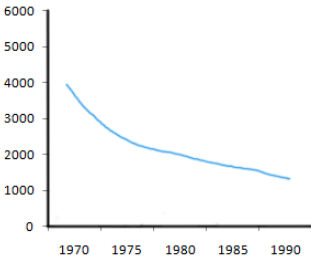
The graph/table/pie chart/bar chart/diagram ...

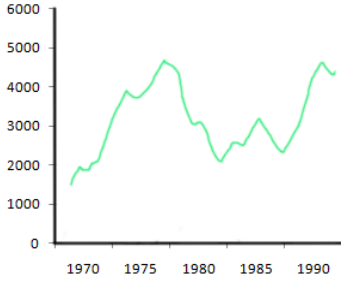
- gives information about/on ...
- provides information about/on ...
- shows ...
- illustrates ...
- compares ...
- explains why ...
- describes ...
- draws the conclusion of (a survey) ...

→ **e.g.** *The pie charts provide information on the proportion of males and females working in agricultural sector.*

TYPES OF CHANGES







NOUNS

a rise (in)* an increase (in) a growth (in) a peak (in) a surge (in) → **e.g.** *a rise in prices*	a fall (in)* a decrease (in) a decline (in) a dip (in) → **e.g.** *a fall in prices*	a fluctuation (in) a variation (in) → **e.g.** *a fluctuation in prices*

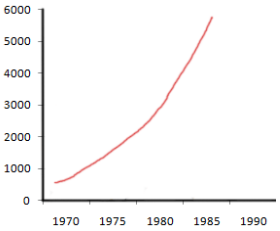
VERBS

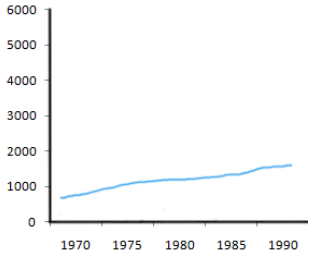
Rises:	*Falls:*	*Fluctuations:*
to rise (by)* to increase to grow → **e.g.** *Prices rose by 3%.*	to fall (by) to decrease to decline to dip (to)* → **e.g.** *Prices fell by 3%.* *Prices dipped to £3.*	to fluctuate(between…and)* to vary (from…to)* → **e.g.** *Prices vary from $2 to $3*

Large rises:	*Large falls:*	*Other:*
to rocket to soar to leap (->leapt) to peak (at)* to surge	to dive to plunge to plummet	To stand (at)* To start (at) To finish (at) To stabilize (at) → **e.g.** *The price of a croissant stood at 2 euros in 2024.*

* Several prepositions can be used in different trend description contexts. Learn more about the use of prepositions in trend description on this page.

DESCRIPTION OF CHANGES





ADVERBS

sharply suddenly rapidly abruptly dramatically significantly considerably markedly wildly	slightly gently gradually steadily modestly marginally
→ **e.g.** *Prices rose sharply*	→ **e.g.** *Prices increased modestly*

ADJECTIVES

sharp sudden rapid abrupt dramatic steep significant considerable marked substantial spectacular	slight gentle gradual steady consistent modest marginal
→ **e.g.** *there was considerable growth*	→ **e.g.** *there was a gradual decline*

USEFUL PHRASES

a small fraction, a small number, a small minority

a large portion, a significant majority

nearly a fifth, almost 10%, in region of 40%, more than a half, over a quarter, around two thirds, more or less three quarters, exactly one in ten, approximately a third

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